

CHEMICAL BONDING

Whenever atoms or ions are strongly attached to one another, we say that there is a *chemical bond* between them. "A strong attractive force that exists between atoms in a molecule is called a **chemical bond**." In other words, "A **chemical bond** is defined as a force that holds two or more atoms together as a stable molecule". Electrons play a key role in chemical bonding.

2.1 Electronic theory of Chemical Bonding

G.N. Lewis and W. Kossel, working independently, used the knowledge of electronic configuration to explain *why atoms joined to form molecules*. They visualised that noble gas atoms had a stable electronic configuration, ns^2np^6 , except He which is $1s^2$, while atoms of all other elements had unstable or incomplete valence shell electronic configuration. All atoms have a tendency to achieve a stable noble gas electronic configuration. In 1916, they put forward the electronic theory of chemical bonding. It states that *in chemical bond formation atoms interact by losing, gaining or sharing of electrons so as to achieve a stable noble gas configuration*. Thus molecules are formed by redistribution of electrons between two or more atoms. We say that atoms react to acquire two electrons, 'duplet' configuration or eight electrons 'octet' structure. H, Li and Be atoms tend to acquire a duplet and all other atoms tend to attain the 'octet' configuration.

Energetics of Chemical Bond Formation

When atoms combine to form molecules, they come closer to each other and interact in such a way that the potential energy of the system is lowered (made more negative) due to the attractive forces. *The lower the potential energy of a system, the more stable it is*. When the potential energy is lowered by 40 kJ/mol or more, we say that atoms are held together by a chemical bond. In order to explain the decrease in energy of the system in bond formation, a graph is plotted between the potential energy of two atoms against the internuclear distance 'R' as shown in Fig. 2.1.

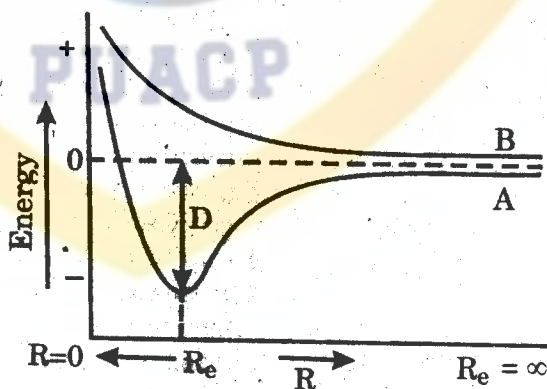


Fig.2.1 Potential energy of two atoms at various distance apart.

When two atoms approach each other, one possibility is that they repel each other. The repulsion increases with decrease of distance. This will result in an increase in the potential energy of the system as shown by curve B. The system will be unstable and the atoms will tend to move away from each other toward an infinite

separation to attain lower potential energy. The other possibility is that the two atoms attract each other as shown by the curve A. The potential energy of the system decreases with the decrease in the distance between them. However, after a certain limits, the forces of repulsion again begin to dominate resulting in a sharp increase of the potential energy. The potential energy is minimum at an equilibrium bond distance ' R_e ' where the two atoms are held together very firmly with the maximum possible decrease in energy. At this ' R_e ' the molecule attains maximum stability and a chemical bond is formed. At distances shorter or longer than ' R_e ' the energy will be higher and the molecule will be unstable.

2.2 Types of Chemical Bonds

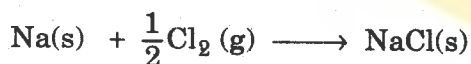
There are five types of bonds recognised by chemists:

1. Ionic or Electrovalent bond
2. Covalent bond or Electron pair bond
3. Coordinate covalent bond
4. Metallic bond
5. van der Waals and long-rang bonds

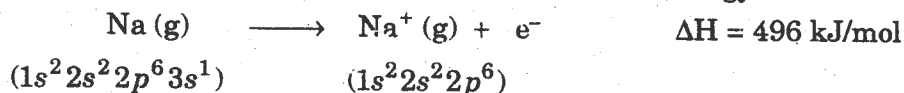
2.3 Ionic or Electrovalent Bond

An ionic bond is the result of electrostatic attraction between positively charged cations and negatively charged anions, which can be formed by the transfer of one or more electrons from one atom or group of atoms to another. *"The electrostatic attraction between the cation and anion produced by electron transfer from one atom to another atom is called ionic or electrovalent bond"*. One atom loses one or more electrons to form a cation and the other gains one or more electrons to become an anion so as to attain the nearest noble gas electronic configuration. Ionic bonding occurs most easily when elements with low ionization energies (metals) react with elements with high electronegativities and high electron affinities (nonmetals). These conditions are generally met when elements from Group IA or IIA react with elements from Group VI A or VII A. Reactions between metals and nonmetals often result in formation of ionic compounds.

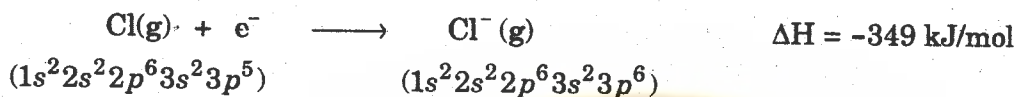
Consider the reaction of sodium (a Group I A metal) with chlorine (a Group VII A nonmetal) to form sodium chloride.



In this reaction, we can picture three major transformations: Sodium atom loses one electron to form sodium ions, Na^+ , to attain the noble gas 'neon' electronic configuration. The energy required for the formation of Na^+ ions from Na in the gas phase is 496 kJ/mol which is simply the first ionization energy of Na.



The chlorine atom gains an electron to form chloride ions to attain the noble gas 'argon' electronic configuration, and releases energy 349 kJ/mol (electron affinity of Cl)



Yet we have described these two processes as occurring simultaneously. The electron is removed from the sodium atom and is transferred to the chlorine atom.

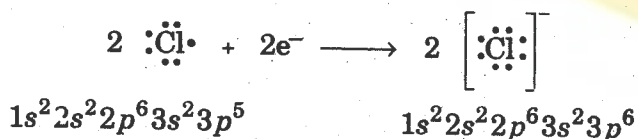
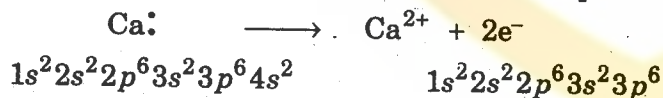
It is often said that NaCl is a stable compound because both the Na^+ and Cl^- ions in NaCl have the stable electronic configuration of a noble gas. However, the energy data cited above show that the completion of noble gas shells cannot account for the stability of NaCl. We find that the process in which an electron is transferred from a gaseous Na atom to a gaseous Cl atom requires an input of energy 147 kJ/mol. The energy gained by the Cl atom is less than the energy cost of removing an electron from a Na atom. Such an unfavourable energy balance is not unique to NaCl. In fact, there is always an unfavourable energy balance in the formation of a pair of ions. Thus, an input of energy is always needed to form an anion and a cation from a pair of elements. Therefore, two separated ions, Na^+ and Cl^- ions, come together, the mutual electrostatic attraction between the ions so formed, releases energy 585 kJ/mol and thus an ionic bond is established. This energy release is more than offsets the 147 kJ/mol required to form the ions. The release of energy results from the electrostatic-(coulombic) attraction of the oppositely charged ions when the distance between their nuclei is reduced to about 0.24 nm in NaCl(g). The Na^+ cation and Cl^- anion do not approach any closer because of coulombic repulsions between the inner-shell electrons.



The overall reaction can be represented as



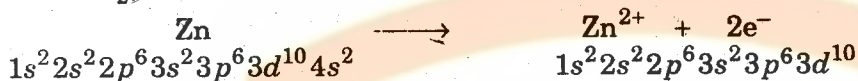
Similarly, formation of calcium chloride can be represented as:



In both the compounds as described above, the positive and negative ions attain noble gas configuration. However, in many ionic compounds the cations do not attain noble gas **configuration** (an octet configuration) as described below:

(i) For some elements (Cu, Ag, Au, Zn, Cd, Hg, In and Tl) all outer shell electrons

are ionized. The ion produced (Cu^+ , Ag^+ , Au^+ , Zn^{2+} , Cd^{2+} , Hg^{2+} , In^{3+} , Tl^{3+}) has a outer shell with 18-electron configuration ($ns^2 np^6 nd^{10}$). For example, Zn^{2+} ion in ZnCl_2 ,



- (ii) For a few post-transition elements (In, Tl, Sn, Pb, Sb, Bi) all outer-shell electrons except the 2s electrons (an "inert pair") are ionized producing ions (In^+ , Tl^+ , Sn^{2+} , Pb^{2+} , Sb^{3+} , Bi^{3+}) with the "18 + 2" electron configuration, $(n-1)s^2 p^6 d^{10} ns^2$.
- (iii) In ion formation among some transition metals, the outer shell s electrons and some number of d electrons are ionized producing the configuration as given below: These metals form ions with variable valency.

Ion	Configuration	Ion	Configuration
Cr^{2+}	$[\text{Ar}] 3d^4$	Fe^{3+}	$[\text{Ar}] 3d^5$
Cr^{3+}	$[\text{Ar}] 3d^3$	Co^{2+}	$[\text{Ar}] 3d^7$
Mn^{2+}	$[\text{Ar}] 3d^5$	Co^{3+}	$[\text{Ar}] 3d^6$
Mn^{3+}	$[\text{Ar}] 3d^4$	Ni^{2+}	$[\text{Ar}] 3d^8$
Fe^{2+}	$[\text{Ar}] 3d^6$	Ni^{3+}	$[\text{Ar}] 3d^7$

We conclude, that there are four types of stable metal ions, namely, the noble gas configuration, the 18-electron configuration, $(n-1)s^2 p^6 d^{10} ns^2$ configuration and the transition metal ion structure.

Ionic Crystal Structures

So far, we have assumed that ionic compounds exist as ion pairs in the gas phase. In fact, they do not. Under normal conditions, ionic compounds are solids. Since electrostatic attraction between oppositely charged ions is independent of direction. Therefore, the ions will be in contact to a number of oppositely charged ions around them in all directions. Like-charged ions, because of their mutual repulsions generally are not in direct contact. Therefore, in solid state, single ionic molecules do not exist in the gas phase. Rather many positive and negative ions are arranged orderly in an alternating cation-anion pattern in three-dimensional arrangement called the **crystal lattice**. When an ionic lattice forms from isolated gaseous ions, energy is released. The number of nearest neighbours of each ion in a lattice is called **coordination number**. X-ray diffraction studies show that a crystal lattice can be represented as made up spherical ions. The relative number of the cations and anions and their sizes play an important role in establishing the geometry of the crystal structure. In order to simplify the job of describing a crystal lattice, it is useful to specify the unit cell of that lattice. The unit cell chosen must;

1. generate the entire crystal by simple translation in three dimensions.
2. indicate the crystal coordination numbers of the ions.
3. be consistent with the formula of the compound.

The smallest structural unit (which can be atoms, ions or molecules) that is repeated in three-dimensional pattern to form a crystal lattice is called the **unit cell**. The "unit cell" is defined by the lengths of edges and the angles between them. There are seven types of unit cells known: cubic, tetragonal, orthorhombic, monoclinic, triclinic, hexagonal, rhombohedral.

Ionic compounds can be classified according to stoichiometric proportions of the cations and anions as 1:1, 1:2, etc and compounds with each classification often have crystallographic in common.

2.4. Structure of Ionic Compounds (NaCl, CsCl, ZnS) of the type 1:1

Four structures account for a large number of the 1:1 ionic solids and these are NaCl, CsCl, zinc blende and the wurtzite. The mineral zinc blende and wurtzite both consist of ZnS, but they have different structures.

Sodium Chloride Structure

An "ion pair" of sodium chloride can be represented as shown in Fig. 2.2(a). When a second ion-pair is brought near to the first, the two will attract each other and arrange themselves as shown in Fig. 2.2 (b). In the $[\text{Na}^+\text{Cl}^-]$ configuration there is an additional attractive force of each Na^+ ion for the Cl^- ion of the other pair. The attractive forces between the oppositely charged ions are much greater than the repulsive forces between the like ions (Na^+ and Na^+ or Cl^- and Cl^-). As a result of this the energy of the system would decrease.

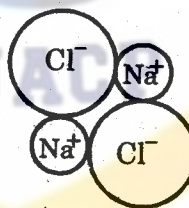
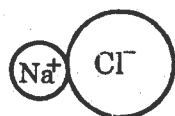
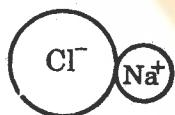


Fig. 2.2(a). An ion pair

Fig. 2.2(b). Two ion-pairs of NaCl

If more than two 'ion pairs' come together, a three dimensional lattice is built up. If four 'ion-pairs' come together, a cubic arrangement is formed (Fig. 2.3). It constitutes the unit cell for NaCl. The three dimensional model of NaCl crystal based upon the unit cell (Fig. 2.3) is shown in Fig. 2.4. The ratio of the ionic radii in the NaCl is 0.53, which suggests an octahedral arrangement in a cubic closest packed structure. Since an octahedral site is formed by six spheres, each Na^+ ion (small sphere) is surrounded by six Cl^- ions (large spheres). In turn, each Cl^- ion is surrounded by six Na^+ ions. We can see that the coordination number of Na^+ is 6 and that this ion sits in an **octahedral hole** defined by the six nearest Cl^- ions.

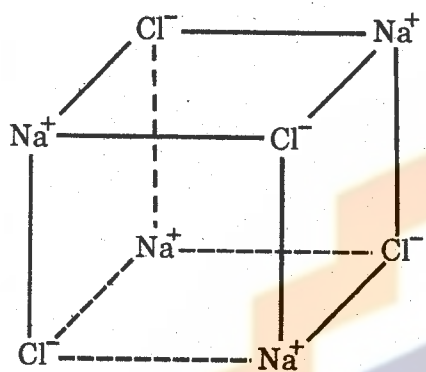


Fig. 2.3: Four 'ion pairs' of NaCl

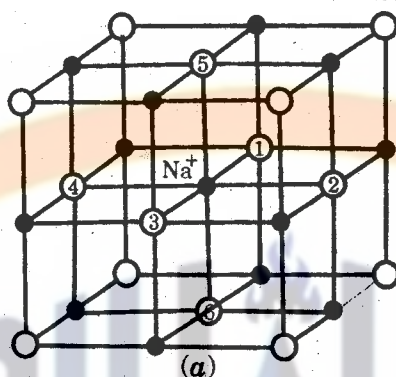
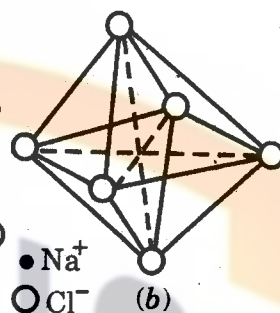


Fig. 2.4: (a) Sodium chloride structure (b) Octahedral Coordination in NaCl



The crystal structure of NaCl is face-centered cubic, which consists of chloride ions at the corners and middles of the faces, and Na^+ ions occupying spaces on the edges and a space at the centre. The Na^+ ion at the centre of the unit cell is surrounded by six chloride ions. Each Cl^- ion in a corner position is shared by eight unit cells and each Cl^- ion in the middle of a face by two unit cells. This leads to a total number Cl^- ions for a unit cell $(8 \times \frac{1}{8}) + (6 \times \frac{1}{2}) = 1 + 3 = 4$. There are 12 Na^+ ions along edges of the unit cell and edge is shared by four unit cells. The Na^+ in the very centre of the unit cell entirely belongs to that unit cell. Thus total number of Na^+ ions for a unit cell $(12 \times \frac{1}{4}) + (1 \times 1) = 3 + 1 = 4$. It is found that a unit cell consists of four Na^+ and four Cl^- ions, which corresponds to stoichiometry of 1:1 keeping with the formula NaCl.

In most of the alkali halides, where the anions are larger than the cations, we find the sodium chloride geometry, with a coordination number of 6. Most of the oxides, sulphides and selenides of alkaline earth metals, AgF, AgCl, AgBr have the sodium chloride structure.

Cesium Chloride Structure

When the cations and the anions are nearly the same size as in CsCl, CsBr or CsI, the ions form the cesium chloride structure, with its coordination number of 8. In CsCl, the radius ratio is 0.93, which suggests a body centred cubic arrangement where each Cs^+ ion is surrounded by 8 Cl^- ions and *vice versa* (Fig. 2.5). The eight neighbouring Cl^- ions are at the corners of a cube surrounding Cs^+ , so this cation is said to occupy a *cubic hole*. It is not reasonable to refer CsCl as body centred cube because this term implies that the body centre is occupied by the same kind of atom as that at the corners of the cube.

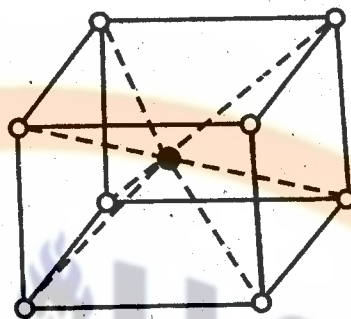
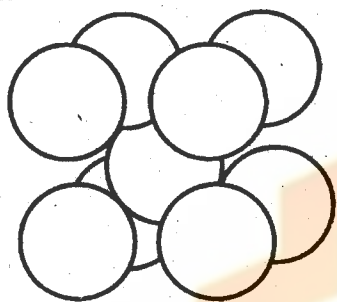


Fig. 2.5(a) The arrangement of ions in CsCl.

(b) A unit cell of CsCl structure.

Structure of ZnS (Zinc blende and Wurtzite)

In ZnS, the great difference in size between the Zn^{2+} cation and the S^{2-} anion leads to a coordination number of 4. The radius ratio for ZnS is 0.40 which suggests a tetrahedral arrangement. The sulphide ions in the ZnS (or zinc blende) lattice form a face-centered cubic arrangement (Fig. 2.6). Each Zn^{2+} ion is surrounded tetrahedrally by four S^{2-} ions. So Zn^{2+} has a coordination number of 4 and sits in a *tetrahedral hole*. ZnO, ZnSe, ZnTe, CdS, CdSe, CdTe, HgS, HgSe, AlAs, AlSb, SiC have the zinc blende structure.

In the *wurtzite structure*, the S^{2-} ions are in hexagonal close packing and Zn^{2+} are in half the tetrahedral holes (Fig. 2.7). ZnO, ZnSe, ZnTe, CdS, CdSe, AlN, BeO form wurtzite structure.

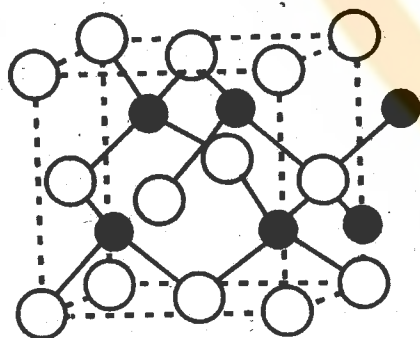


Fig. 2.6 Zinc blende (ZnS)

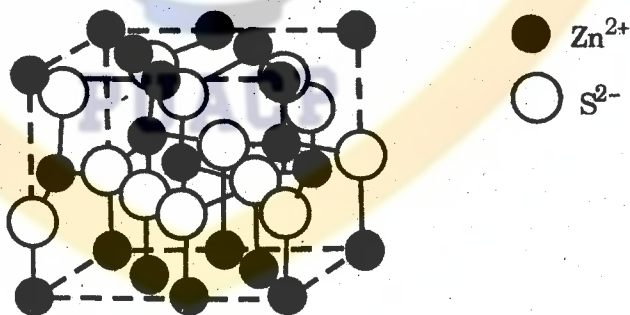


Fig. 2.7 Wurtzite (ZnS)

Ionic Compounds of the type 1:2 (CaF_2 , TiO_2)

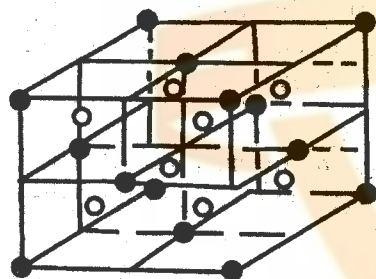
The crystal structures for the ionic compounds of the type 1:2 are more complex. The crystals have two coordination numbers, one for the cation and another for the anion, because the cations and anions occur in unequal numbers.

In CaF_2 (the fluorite structure) there are twice as many F^- as Ca^{2+} ions. The fluorite structure is a cubic close-packed (ccp) lattice of positive ions (Ca^{2+}) with negative ions occupying all the tetrahedral holes. Each Ca^{2+} ion is surrounded by eight F^- ions at the corners of a cube and each F^- ion is surrounded tetrahedrally

by four Ca^{2+} ions. Thus the crystal coordination number of Ca^{2+} is eight, that of F^- is four. The fluorite structure is formed when the radius ratio is 0.73 or above. Compounds exhibiting the fluorite structure are of BaF_2 , SrCl_2 , ThO_2 .

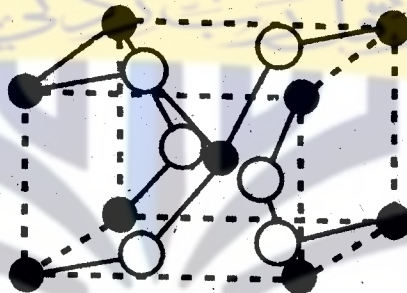
When the positions of the positive and negative ions are reversed from those in the fluorite structure, the result is the **antifluorite structure**. In the crystal lattice of Li_2O , for example, Li^+ ions fill all the tetrahedral holes in a ccp lattice of O^{2-} ions. Compounds exhibiting the antifluorite structure are oxides, sulphides of Li, Na and K.

The **rutile structure** (TiO_2), is formed usually where the radius ratio is between 0.43 to 0.73. In TiO_2 , Ti^{4+} has a crystal co-ordination number of six and O^{2-} , three. In this structure, two of the O^{2-} ions are within the interior of the cell, two are in the top face, and two in the bottom face of the cell. Ti^{4+} ions are at the corners and the centre of the cell (Fig. 2.9). The rutile structure is adopted by many metallic oxides, eg. PbO_2 , MoO_2 and MnO_2 .



● Calcium = Ca^{2+}
○ Fluoride = F^-

Fig. 2.8. The fluorite (CaF_2) structure.



● = Titanium = Ti^{4+}
○ = Oxygen = O^{2-}

Fig. 2.9. The rutile (TiO_2) structure.

2.5 Radius Ratio

The structures of solids are determined by the relative numbers of different ions and their sizes. The relative size of different ions is expressed in terms of the **radius ratio**. The radius ratio is the ratio of the radii of the cations and the anions.

$$\text{Radius} = \frac{r_+}{r_-} = \frac{\text{radius of cation}}{\text{radius of anion}}$$

In general, the radius ratio can be expressed as

$$\text{Radius} = \frac{\text{radius of central sphere}}{\text{radius of sphere surrounding it}}$$

Limiting radius ratio is the maximum ratio of the radius of the cation to the radius of the anion upto which the crystal structure is retained. Above this radius ratio the particular crystal structure will not be retained. The coordination numbers for the various radius ratios have been calculated and are given in table 2.1 along with the expected structure.

Table 2.1. Value for the Radius Ratio

No.	Structure	Coordination number	Limiting ratio, r_+/r_-
1	Body centered cubic	6	0.73 and above
2	Octahedral	6	0.414- 0.73
3	Square planar	4	0.414- 0.73
4	Tetrahedral	4	0.225- 0.414
5	Triangular	3	0.155- 0.225
6	Linear	2	0.155

2.6 Lattice Energy (Crystal Energy)

"The lattice energy is the amount of energy evolved when gaseous ions of opposite charges are brought together to form an ionic crystal composed of 1 mole of formula units of a compound". Alternatively, lattice energy is defined as the amount of energy required to convert one mole of an ionic solid to its gaseous ions. The lattice energy depends upon the strength of the interactions between the particles in the solid and the geometry of the crystal structure. It is a measure of the stability of an ionic solid. The greater the lattice energy, the stronger the bonds. Lattice energies may be determined in two different ways; by theoretical calculation or by experimental measurement (i.e., Born-Haber cycle).

The theoretical approach involves the calculation of interaction (attractive and repulsive) energies among all the particles in a crystal. The calculation gives a result which is simply represented as follows;

$$\text{Lattice energy, } U = A \frac{q^+ q^-}{r_+ + r_-} - B$$

Where A is a constant whose value depends upon the kind of crystal lattice, q^+ and q^- are the charges on the ions, r_+ and r_- are the ionic radii and B is a small constant of repulsion. The important quantities which determine the magnitude of the lattice energy are, therefore, ionic charges q , and the ionic radii. Equation shows, that the crystal lattice energy increases in magnitude as the charges on the ions increase. Thus most ionic solids containing multiply charged ions have higher crystal lattice energies than those containing only monovalent ions. The equation also shows that the crystal lattice energy gets larger as the separation between cation and anion decreases, that is as r_+ and r_- decrease.

2.7 Characteristics of Ionic Compounds

The ionic compounds are made of positive and negative ions held by electrostatic forces in a crystal lattice. Each ion is surrounded by the opposite ions in alternate positions in a definite order in all directions. The strong coulomb forces existing between ions in ionic compounds render special properties to them and some of these are discussed below:

1. **Solids at room temperature.** On account of strong electrostatic forces between the opposite ions, these ions are locked in their allotted positions in the crystal lattice. Since they lack the freedom of movement, therefore, they

are solids at room temperature.

2. **High Melting and Boiling points.** Ionic compounds have high melting and boiling points because the ions are tightly held in the lattice by strong attractive forces and high thermal energy is needed to separate them from one another. Therefore, ionic compounds tend to have high melting and boiling points.
3. **Hard and Brittle.** Ionic crystals are hard and brittle. Their hardness is due to the strong electrostatic forces which hold each ion in its allotted position.

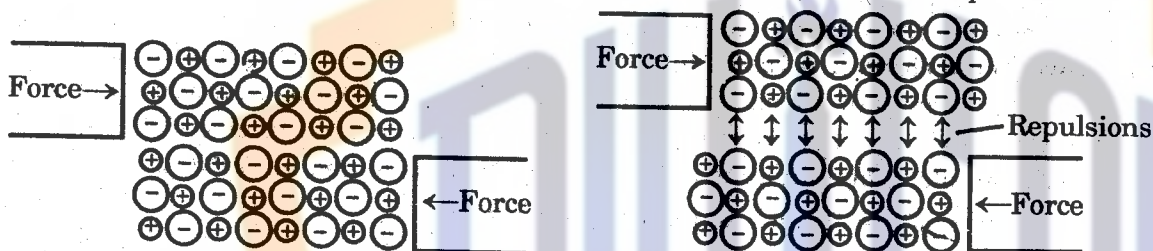


Fig. 2.11 Brittleness in ionic solids

Ionic crystal consists of layers of (+) and (-) ions in alternate positions so that the opposite ions in the various parallel layers lie over each other. When an external force is applied to an ionic crystal to start one layer of ions moving across the next, a slight shift brings the like ions in front of each other and thus interionic repulsions cause the two layers to separate. The crystal cleaves here.

4. **Electrical Conductivity.** Ionic crystals are poor conductors of electricity because the ions are not free to move about. However, in the molten state or dissolving in polar solvents, the ions become quite free to migrate in an electric field and conduct electricity.

5. **Solubility.** Ionic compounds are generally quite soluble in polar solvents like water. Since dissolution involves the rupture of crystal lattice and separation of ions which is enhanced by polar solvents only. Nonpolar solvents do not affect the ionic crystal lattice and would not be able to dissolve the ionic compounds. When a crystal of an ionic compound is placed in

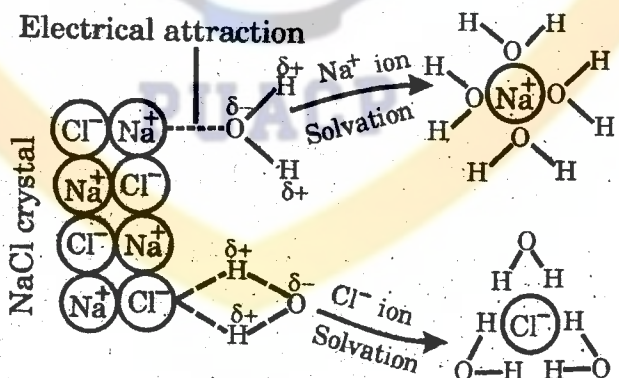
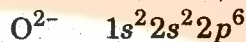
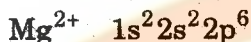
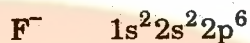
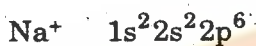


Fig. 2.12 Solvation of NaCl in water

water, the polar water molecules detach the (+) and (-) ions from the crystal lattice by their electrostatic pull. These ions then get surrounded by water molecules and can lead an independent existence and are thus dissolved in water.

6. **Crystalline State.** Ionic compounds are composed of cluster of ions in the crystal lattice and form definite crystal structure.
7. **Isomorphism.** Ionic compounds with similar constitution give similar crystal structure which have identical electronic configuration of ions and hence they are

isomorphous. For example, NaF is isomorphous with MgO. Both possess similar crystalline structure and similar electronic configuration.



8. **Non-directional Bonds.** The ionic bonds are held together by electrostatic force of attraction between oppositely charged ions. These forces are non-directional, therefore, ionic bond is non-directional.
9. **Ionic reactions are fast.** Most of the reactions of ionic compounds in solution are almost instantaneous, since in solution the ions exist as individual units (of course solvated). The reacting ions have simply to collide to give the products. For example, when an aqueous solution of NaCl is added to AgNO₃ solution, a white precipitate of AgCl is immediately formed.



Variable Valency

Certain elements exist in more than one oxidation states and thus show variable valency. Variable valency of an element may be due to different reasons as discussed below:

1. In transition elements the variable valency is due to the involvement of 'd' electrons. The elements show a wide range of oxidation states depending upon the number of valency electron involved. For example, iron (Fe) shows normally Fe²⁺ and Fe³⁺ oxidation states.
2. Some of the 'p' block elements show variable valency due to the phenomenon of "inert-pair" electrons. In the valency shell of heavy metals the two s electrons remain inert and only the 'p' electrons are lost to form the +ve ions. Thus an element shows higher and a lower oxidation state. For example, tin (Sn) shows Sn²⁺ and Sn⁴⁺ oxidation states.
3. The 'p' block elements which have empty 'd' orbitals can show variable valency by utilizing these 'd' orbitals in addition to valency shells s and p orbitals. For example, phosphorus (1s²2s²2p⁶ 3s²3p³ 3d⁰) can form trivalent and pentavalent halides e.g., PCl₃ and PCl₅. In PCl₅, 3d orbitals of phosphorus are involved.

2.8 Covalent Bond

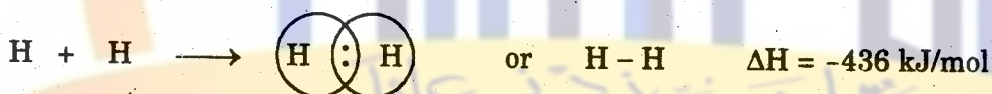
A chemical bond formed by the mutual sharing of electrons between two atoms is called a **covalent bond**. Each atom contributes one electron of its valence shell to the shared electron pair. This pairing of electrons according to Lewis, should lead to a noble gas configuration. The mutually shared pair of electrons is considered to be possessed by both the atoms in common, which would contribute to the electronic configuration of each atom in the molecule.

The sharing of electrons by two atoms in a covalent bond is energetically favourable. Energy is liberated when two isolated atoms form a covalent bond. Since the shared electrons tend to lie between the two bonded nuclei, so the shared

electrons are attracted by both nuclei. At the same time, the nuclei of two atoms also repel each other as do the two electrons. The repulsion of the nuclei is offset by this attraction. It is the net attractive force between the shared electrons and the nuclei that holds the atoms together. **"The attractive force between atoms created by sharing of an electron pair is called a covalent bond".**

A shared electron pair can be represented either by two dots or by a line (–) between the two bonded atoms. Although Lewis used only dots, modern structures generally use lines to represent shared electron pairs. Such structures are called *Lewis structures*.

Atoms of the same electronegativities or low electronegativity differences usually form covalent bonds. Reactions between two nonmetals result in covalent bonds. For example, H_2 molecule is made of two H atoms, each having one valence electron. Each contributes an electron to the shared pair and both atoms acquire stable helium configuration, resulting a stable H_2 molecule.



In Cl_2 molecule, each chlorine atom has seven valence electrons (2, 8, 7). Each atom contributes one electron to the shared electron pair and both the chlorine atoms achieve a stable octet. Thus each Cl atom acquires the stable electronic configuration of the noble gas argon.

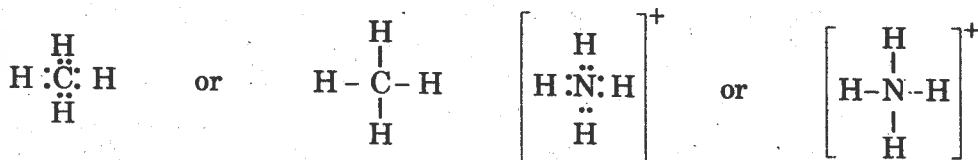


Note that each Cl atom has pairs of electrons that are not involved in bonding. These electrons are called **nonbonding electrons** or **lone pair electrons** or **unshared electron pairs**.

Certain atoms (notably C, N, O and S) may share more than one pair of electrons to achieve an octet. Two atoms form a **single covalent bond** when they share one pair of electrons, a **double covalent bond** when they share two electron pairs, and a **triple covalent bond** when they share three electron pairs.



Lewis dot formulas can also be used to represent atoms covalently bonded in a molecule or a polyatomic ion.



Lewis dot formula

Lewis dot formula (ammonium ion)

The covalent bonds are of two types: polar, and non-polar.

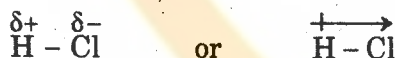
2.9 Polarity of Molecules

The concept of polarity is useful for describing entire molecules as well as bonds. Molecules that have polar bonds tend to have an overall polarity. Such molecules are called polar molecules. However, the polarity of the individual bonds sometimes cancels, and the molecule as a whole is nonpolar, e.g., CO_2 , BF_3 and CCl_4 .

"A covalent bond between two like atoms such as $\text{H}-\text{H}$, $\text{Cl}-\text{Cl}$, in which the electron pair is shared equally between the two nuclei is called a **nonpolar bond**". Stated differently, in nonpolar covalent bonds the electron density is symmetrical about a plane that is perpendicular to a line about the two nuclei. We can generalize that the covalent bonds in all homonuclear diatomic molecules must be nonpolar. Nonpolar compounds include homonuclear diatomic molecules, some binary compounds of nonmetals and most compounds of hydrogen and carbon.

"A covalent bond between two unlike atoms such as $\text{H}-\text{Cl}$, $\text{H}-\text{Br}$, in which the electron pair is shared unequally and the bonded atoms acquire a partial positive and negative charge, is called a **polar covalent bond**". The polarity of a bond, that is, the degree to which an electron pair is unequally shared, depends on the difference between electronegativities of the two bonded atoms. The greater the electronegativity difference, greater the polarity.

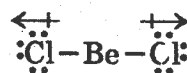
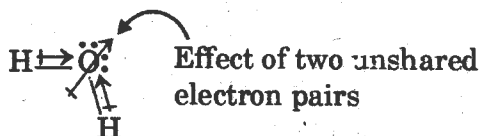
A molecule having partial positive and negative charge separated by a distance is commonly referred to as a Dipole (two poles). The dipole of a bond is indicated by an arrow from positive to negative end with a crossed tail as shown in HCl molecule.



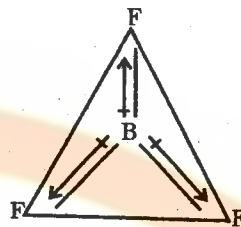
In order for a molecule to be polar, both of the following conditions must be met:

1. There must be at least one polar bond or one lone pair on the central atom.
2. The polar bonds, if there are more than one, must not be so symmetrically arranged that their bond polarities cancel.

Consider as example, the H_2O molecule and the BeCl_2 molecule. The water molecule is angular, with two lone pairs of electrons on the oxygen atom. The bond dipoles of H_2O are directed from H to O because O is more electronegative than H. These are reinforced by strong dipoles associated with the two lone pairs on the O atom. The result is that the H_2O molecules are very polar. The BeCl_2 molecule is linear. Each $\text{Be}-\text{Cl}$ bond is polar. However, the bond dipoles are identical in magnitude and opposite in direction, and therefore, they cancel to give nonpolar molecules



BF_3 has polar bonds but the BF_3 molecule is nonpolar, whose geometry is trigonal planar. The three bond moments are of equal magnitude. They point to the corners of an equilateral triangle. Because of the symmetry of the molecule, the bond moments cancel.



Molecules with symmetrical structures, but different atoms bonded to the central atom, are usually polar. For example, tetrahedral CH_4 molecules are nonpolar, however, tetrahedral CH_3Cl molecules are polar.

2.10 Theories of Chemical Bonding

The bonding theories that are currently accepted allow us to predict structures and properties that are usually accurate (although they are not always entirely satisfactory). As always, we should keep in mind the truism that whatever we propose must be consistent with experimentally determined facts. When there is disagreement between facts and theory, theory must be modified to accommodate all known facts.

We shall discuss three theories. The first is the Valence Shell Electron Pair Repulsion (VSEPR) theory, which assumes that electron pairs are arranged around the central atom in such a way that there is maximum separation (and, therefore, minimum repulsion) among electron pairs. The second theory is the Valence Bond (VB) theory, which explains the bonding in terms of overlapping atomic orbitals. The third theory, is called Molecular Orbital (MO) theory, which assumes that the atomic orbitals of the original unbonded atoms become replaced by a new set of molecular energy levels, called *molecular orbitals*, and the occupancy of these orbitals determines properties of the resulting molecule.

2.11 Valence Shell Electron Pair Repulsion (VSEPR) Theory

Lewis structures provide a very useful method for representing the structures of covalent molecules. Lewis structures, however, do not give any information about the actual geometry (shape) of the molecule. Sidgwick and Powell in 1940 pointed out that the shapes of the molecules can be predicted on the basis of electron pairs present in the valence shell of the central atom. The VSEPR theory explains observed geometry of molecules and polyatomic ions by an electrostatic picture based primarily on the Lewis structure, without reference to orbitals. The main idea of the VSEPR theory is that:

The electron pairs (both lone pairs and shared pairs) surrounding the central atom must be arranged in space as far apart as possible to minimise the electrostatic repulsion between them.

A **central atom** is any atom that is bonded to two or more than two other atoms. The first and the most important rule of the VSEPR theory is that *the bond angles about a central atom are those that minimize the total repulsion between the electron pairs in the valence shell of the atom*. While working out the shapes of

molecules from this theory, the following rules must be remembered:

1. *A lone pair of electrons occupies more space than a bonding pair.* This is because the lone pair of electrons is under the influence of only one nucleus of the central atom, they are expected to occupy more space with a greater electron density than the bond pair electrons which are under the influence of two nuclei. Thus decreasing order of repulsion is:
lone pair-lone pair > lone pair-bond pair > bond pair-bond pair
2. *Repulsive forces decrease sharply with increasing interpair angle.* They are strong at 90° , much weaker at 120° , and very weak at 180° .
3. *The influence of a bonding electron pair decreases with the increasing value of electronegativity of an atom forming a molecule.*
4. *Multiple bonds behave as a single electron-pair for the purpose of VSEPR theory.*
5. *The two electron pairs of a double bond (or three electron pairs of a triple bond) occupy more space than the one electron pair of a single bond.*
6. *The lone pairs repel bond pairs giving rise to some distortions in the molecular shape. The distortion may also result due to different atoms in the molecule.*

Applications of VSEPR Theory

Let us now apply the valence shell electron pair repulsion theory to predict the shapes of molecules. The first step in the VSEPR method for determining the geometry of the molecule is to write down its Lewis structure in order to determine the number of electron pairs around the central atom. The second step is to determine the total number of electron pairs (lone and bonding) around the central atom. The third step is to determine the number and location of lone pairs. The molecules can be classified according to the number of electron pairs around the central atom.

Gillespie proposed the following rules to explain the shapes of inorganic molecules.

Rule 1. *If the central atom of a molecule is surrounded only by bonding electron pairs (bp's) and not by non-bonding electron pair called lone pairs (lp's), the geometry of the molecule will be regular, i.e., it will be linear, triangular planar, tetrahedral, trigonal bipyramid and regular octahedron for 2, 3, 4, 5, and 6 bonding electron pairs.*

Rule 2. *When the central atom in a molecule is surrounded by both bp's and lp's, the molecule does not have a regular shape. The alternation or distortion in shape is due to the alternation in bond angles which arises due to the presence of lp's on the central atom.*

Rule 3. *B-A-B bond angle decreases with the increase in electronegativity of atom B in AB molecule wherein A is the central atom. This is due to the fact that with the increase in electronegativity of atom B, the average position of the bonding electron-pair moves farther from the central atom A and hence the bond angle decreases.*

exerted by bonding electron-pair on the electron-pair on atom A decreases. For example, PI_3 (102°) $>$ PBr_3 (101.5°) $>$ PCl_3 (100°)

Rule 4. Bond angles involving multiple bonds are generally larger than those involving only single bonds. However, the multiple bonds do not affect the geometry of a molecule.

Rule 5. Repulsion between electron-pairs in filled shells are larger than the repulsion between electron pairs in incompletely filled shells, e.g. compare H–O–H and H–S–H bond angles in H_2O and H_2S molecules: H_2O (105.50°) $>>$ H_2S (92.20°)

The **valence shell electron pair repulsion (VSEPR) model** predicts the shapes of molecules and ions in which valence-shell electron pairs are arranged about each atom so that electron pairs are kept as far away from one another as possible, thus minimizing electron-pair repulsions. **Molecular geometry** is the general shape of a molecule, as determined by the relative positions of the atomic nuclei. Please remember, that even though VSEPR suggests pairs of electrons, VSEPR really refers to electron domains or regions. Remember, an electron region can exist as a single electron (one electron), two electrons (lone pair or bonded pair), four electrons (a double bond), or six electrons (triple bond). However, when we determine the geometry of a molecule, we locate the positions of the atoms, not the electron pairs. For the **General molecular formula**, “A” refers to central atom, “B” refers to atoms attached to central atom, and “E” refers to unbonded electron pairs. For example, AB_2E_2 is shown as the formula of water (H_2O), which has two bonded hydrogen atoms (“B”) and two lone (unbonded) electron pairs (“E”).

1. AB_2 : Molecules with two electron pairs around a central atom

There are several molecules and ions that consists of a central atom plus two atoms of another element in which there are no lone pairs on the central atom. This kind of molecule is abbreviated as AB_2 . Typical compounds include BeCl_2 , BeBr_2 , BeI_2 , and CO_2 , as well as CdX_2 and HgX_2 , where X = Cl, Br or I. A polyatomic ion in this category is the nitronium ion NO_2^+ . All of these are known to be **linear** (bond angle = 180°).

In the solid state, **Beryllium chloride**, BeCl_2 , molecules are bonded to each other in a polymeric solid. However, BeCl_2 exists as discrete molecules in the gaseous state. The Lewis structure for BeCl_2 shows that the central atom Be has two bonding electron pairs and has no lone pairs of valence electrons, so it has two electron pairs. VSEPR theory, which assumes that electron pairs will be as far apart from one another as possible, predicts that the two electron pairs on Be will be 180° apart. Thus VSEPR theory predicts a **linear** structure for BeCl_2 , and for all other molecules of this type.

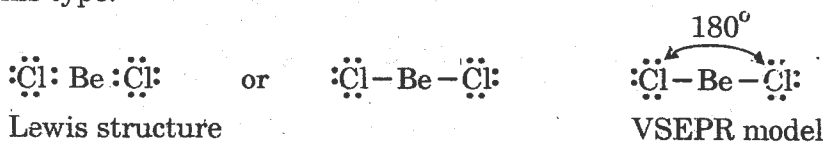


Fig. 2.13. BeCl_2 molecule

The Lewis structure for **nitronium ion**, NO_2^+ , shows that the central nitrogen atom has two bonding electron pairs, since a double bond behaves as a single electron pair-bond in VSEPR model, and has no lone pair of valence electrons. Thus VSEPR theory predicts and experimental observations confirm that the **nitronium ion is linear**, which gives maximum separation between the two electron pairs about the central atom.

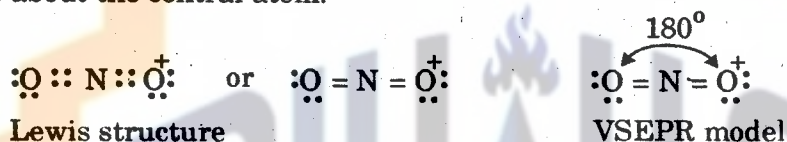


Fig. 2.13 Geometry of NO_2^+ ion

2. Molecules with three electron pairs arounds a central atom

AB_3 ; When the central atom of a molecule is surrounded only by three bonding electron pairs, the geometry of the molecule will be **triangular planar**. Boron forms many covalent compounds by bonding to three other atoms. Typical examples include BF_3 , BCl_3 , BBr_3 and BI_3 . All are **trigonal planar** (that is, the flat molecules in which the bond angles are 120°).

The Lewis structure of boron trifluoride, BF_3 , shows that the central atom B has three bonding electron pairs and has no lone pair of electrons. VSEPR theory predicts a trigonal planar structure for BF_3 molecule because this structure gives maximum separation among the three bonding electron pairs. The structures of BCl_3 , BBr_3 and BI_3 are similar.

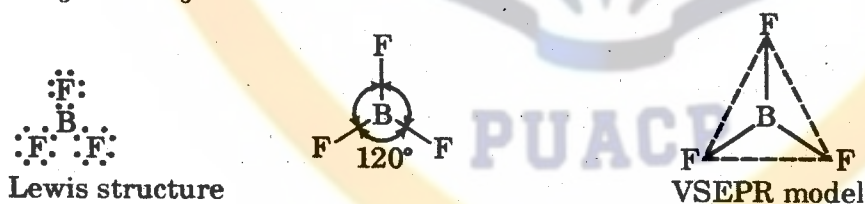
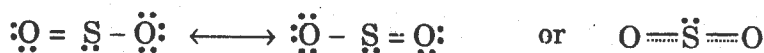


Fig. 2.14 BF_3 molecule; trigonal planar.

AB_2E ; When the central atom of a molecule is surrounded by two bonding electron pairs and by one lone-pair of electrons, the geometry of the molecule will be **bent, angular or V-shaped**, e.g., SO_2 , SnCl_2

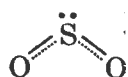
The Lewis formulas for the sulphur dioxide, SO_2 , has two equivalent resonance structures.



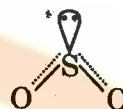
Two equivalent resonance structures

In SO_2 , the central atom sulphur is surrounded by three electron pairs. In SO_2 the resonance form contains a single bond, a double bond, and a lone pair of electrons, but a double bond is counted as a single electron pair for the purpose of VSEPR model. Therefore, in effect, S has three electron pairs around it. The VSEPR

theory predicts a trigonal planar arrangement for the three regions of electron density.



trigonal planar electronic geometry and angular molecular geometry



If all the three regions of electron density about the sulphur atom were equivalent, we would expect the O-S-O bond angle to be 120° . Clearly the three regions of electron density are not equivalent because the two regions are associated with the S-O bonds, while the third region is the lone pair of electrons that reside on the S atom. As the lone pair occupies more space, it will push the two bond pairs closer to each other and the bond angle between two bond pairs becomes less than 120° .

The Lewis structure of stannous chloride, SnCl_2 , shows that the central Sn atom has three electron pairs, out of which one is lone pair. The VSEPR theory predicts a trigonal planar geometry for the three electron pairs. The lone pair of electrons exerts a greater repulsion on the bond pairs resulting in the shortening of Cl-Sn-Cl angle. Thus SnCl_2 in vapour phase is V shaped and arrangement is called angular.

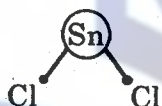
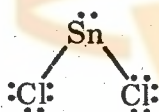


Fig. 2.15. SnCl_2 molecule V shaped

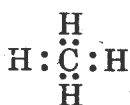
The trigonal planar molecular geometry occurs when there are three atoms bonded and no lone pair of electrons on the central atom and the bond angle is 120° . The angular molecular geometry occurs when there are two atoms bonded and one lone pair of electrons on the central atom and the bond angle is slightly less than 120° .

3. Molecules with four electron pairs around a central atom

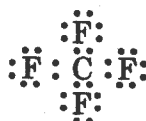
AB_4 ; When the central atom of a molecule is bonded to the four other atoms and has no lone-pair of electrons on it, the molecular geometry will be **tetrahedral**, e.g., CH_4 , CCl_4 , SiF_4 .

The Group IVA elements form numerous covalent compounds by sharing their four electrons with four other atoms and are abbreviated as AB_4 molecules.

The lewis structures for **methane**, CH_4 and **carbon tetrafluoride**, CF_4 are

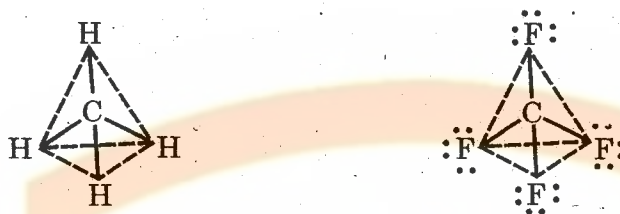


Methane

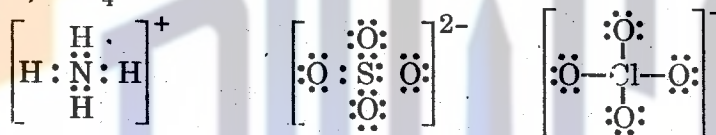


Carbon tetrafluoride

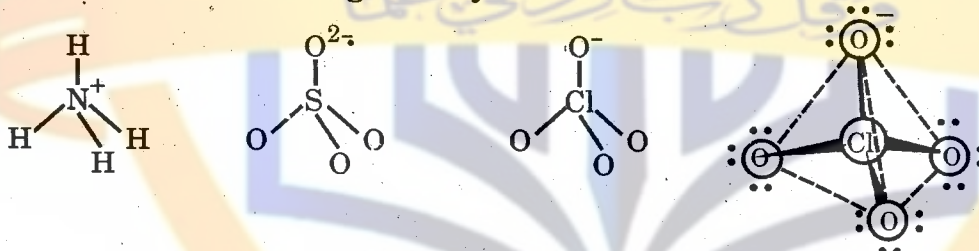
The central carbon atoms in both molecules have four bonding electron pairs, so VSEPR theory predicts **tetrahedral geometry** since this gives maximum separation. The four H atoms in CH_4 and four F atoms in CF_4 , lie at the corners of a regular tetrahedron. The H-C-H bond angles are all $109^\circ 28'$.

Fig. 2.16. Shape of CH_4 and CF_4 (tetrahedral)

The Lewis structure of **Ammonium ions**, NH_4^+ , **sulphate ion**, SO_4^{2-} and **perchlorate ion**, ClO_4^- are shown below:

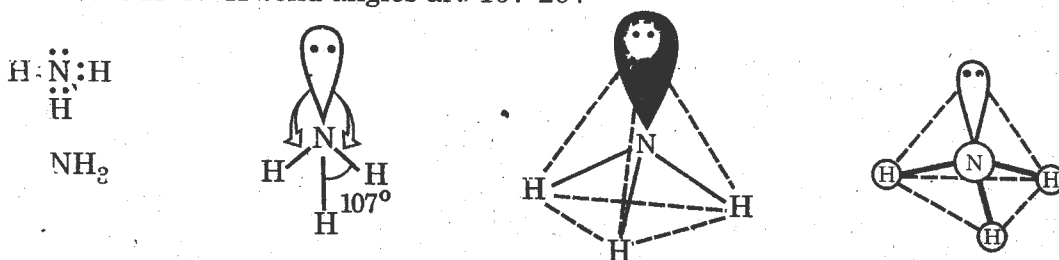


Each has four bonding pair of electrons, which are directed to the corners of a regular tetrahedron for maximum separation from each other. Thus NH_4^+ , SO_4^{2-} and ClO_4^- have **tetrahedral ionic geometry**.



AB_3E ; When the central atom of a molecule has three bonding electron pairs and one lone-pair of electrons on it, the molecular geometry is **trigonal pyramidal**, e.g., NH_3 , PCl_3 .

The Lewis structure for **ammonia** molecule, NH_3 , shows that the central N atom has three bonding electron pairs and one lone electron pair. The VSEPR theory predicts tetrahedral geometry with bond angles of $109^\circ 28'$. The shape of the molecule is determined by the arrangement of atoms and not the lone pair of electrons. Therefore, the NH_3 molecule is described as **pyramidal**. The N atom lies at the apex and the three H atoms lie at the corners of a pyramid with a triangular base. Also, repulsions between the lone pair and the bonding electron pairs are greater than repulsions between bonding electron pairs. The VSEPR theory thus predicts that the H-N-H bond angles will be less than $109^\circ 28'$. Experimentally, we find that the H-N-H bond angles are $107^\circ 20'$.

Fig. 2.17 Molecular geometry of NH_3 , pyramidal

AB₂E₂; When the central atom of a molecule is bonded to the other two atoms and two non-bonding pair on it, the molecular geometry is **angular** or **bent**, e.g., H₂O, SCl₂.

The Lewis structure for **water** molecule, H₂O, shows that the central O atom is surrounded by two bonding electron pairs and two lone electron pairs. For four electron pairs, VSEPR theory predicts tetrahedral geometry, with bond angles of 109°28'. Since there are two lone pairs of electrons, in the H₂O molecule, VSEPR theory satisfactorily explains the **angular molecular structure** of water molecule. The two lone pairs occupy more space than the two bonding pairs and generate greater repulsion between two lone pairs than the repulsion between a lone pair and a bonding pair, which in turn is greater than the repulsion between two bonding pairs. Thus the two O-H bonds in H₂O will be forced to come closer together than N-H bonds in NH₃ and account for the observed bond angle of 104.5° in H-O-H. Typical examples are H₂O, H₂S, N₂O and Cl₂O and all are angular molecules.

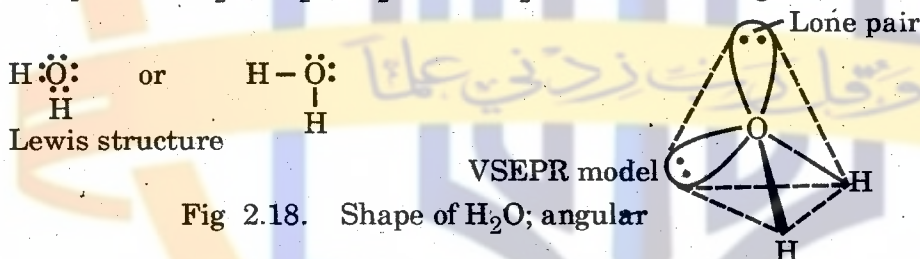


Fig 2.18. Shape of H₂O; angular

Greater the number of lone pairs on a central atom, the greater is the contraction in the bond angle.

4. Compounds with five electron pairs around a central atom

AB₅; When the central atom of a molecule is surrounded only by five bonding electron pairs, the molecular geometry is **trigonal bipyramidal**.

The heavier group VA elements, P, As, Sb, form some covalent compounds by sharing all five of their valence electrons with five other atoms. PF₅ and PCl₅ are such compounds. The Lewis structure for PCl₅ shows five bonding electron pairs around the central P atom. The VSEPR theory predicts **trigonal bipyramidal geometry**, because maximum separation of five bonding electron pairs around a central P atom is achieved when the five Cl atoms lie at the corners and P atom in the centre of a trigonal bipyramid.

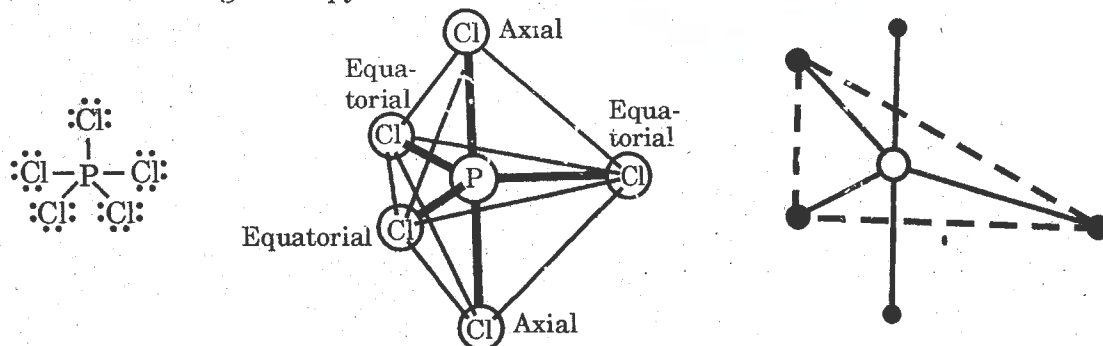


Fig.2.19 PCl₅ molecule, trigonal bipyramid

The repulsion of the five electron pairs is minimized in this arrangement. The five positions and hence the five bonds of a trigonal bipyramid are not all equivalent. Two of the positions are called *axial*. In the PCl_5 molecule they form a linear system. The other three positions are called *equatorial*. They lie in a plane at the corners of an equilateral triangle. The bond angle between the equatorial Cl atoms is 120° , the bond angle between the two axial Cl atoms is 180° , and the bond angle between an axial Cl atom and an equatorial Cl atom is 90° . The axial bonds are somewhat longer than the equatorial bonds in such molecules. The two axial Cl atoms and the three equatorial Cl atoms in PCl_5 can be distinguished experimentally because of the different bond angles and bond lengths.

AB₄E. When the central atom of a molecule is surrounded by four bonding electron pairs and by one lone-pair of electrons, the molecular geometry is **seesaws** (irregular tetrahedral), e.g., SF_4 , TeCl_4 .

The sulphur has five electron pairs around it; four from the S-F bonds and one from the nonbonding pair. The five electron pairs about sulphur should have a trigonal bipyramidal arrangement. Because the axial and equatorial positions of the electron pairs are not equivalent, we must decide in which of these positions the lone pair appears. It is experimentally observed, the nonbonding (lone) pair occupies an equatorial position and the four bonding pairs occupy the remaining four positions. This gives a **seesaw** (or distorted tetrahedral) geometry. The axial and equatorial S-F bonds are slightly bent back away from the nonbonding electron region, suggesting that the bonding regions are "pushed" by the nonbonding regions, which is larger and greater repulsion.

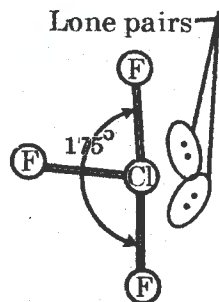
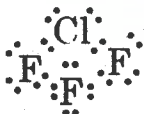


Lewis Structure

Geometry of SF_4

AB₃E₂; When the central atom of a molecule is surrounded by three bonding electron pairs and two lone pair of electrons gives a **T-shaped geometry**, e.g., ClF_3 .

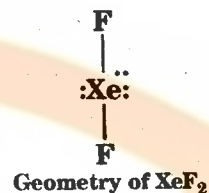
Chlorine Trifluoride, ClF_3 , has five electron pairs around the central Cl atom, two of them are lone pairs and the rest of the three are bonding pairs. The VSEPR theory predicts *trigonal bipyramidal geometry* for the presence of five electron pairs. The two lone-pairs would set up repulsive interactions with the

Fig. 2.20 Molecular geometry of ClF_3 ; T-shaped.

bond-pairs and will distort its symmetry from true trigonal bipyramidal geometry. Experimental evidence indicates that it is more favourable for the lone pairs to be in the equatorial positions. Therefore, two of the three F atoms lie in the axial positions. The four atoms form a distorted "T", and this structure gives minimum

repulsion. One F-Cl-F bond has bond angle 180° and two F-Cl-F have bond angles of 90° each.

AB₂E₃; When the central atom of a molecule is bonded to the two atoms of another element and has three nonbonding pair on it, the molecular geometry is **linear**, e.g., XeF₂. The three lone pairs on Xe occupy the equatorial positions of the trigonal bipyramidal arrangement, giving a linear geometry.



5. Molecules with six electron pairs around a central atom

AB₆; When the central atom of a molecule is surrounded only by six bonding electron pairs and has no bonding pair, the geometry of the molecule is **octahedral**. The halides of the group VIA elements in the +6 oxidation state are examples of the **AB₆ type**. The Lewis structure of **sulphur hexafluoride**, SF₆, shows that there are six bond pairs around the central S atom. The octahedron VSEPR model correctly predicts octahedral geometry, since the maximum separation possible for six F atoms surrounding one S atom is achieved when the F atoms are situated at the corners and the S atom in the centre of a regular octahedron and thus repulsion of the six electron pairs is minimized in this arrangement. An octahedron is a regular solid figure whose corners are all identical. Therefore, all the bonds and all the F atoms are equivalent in SF₆ molecule. The F-S-F bond angles are 90° or 180° .

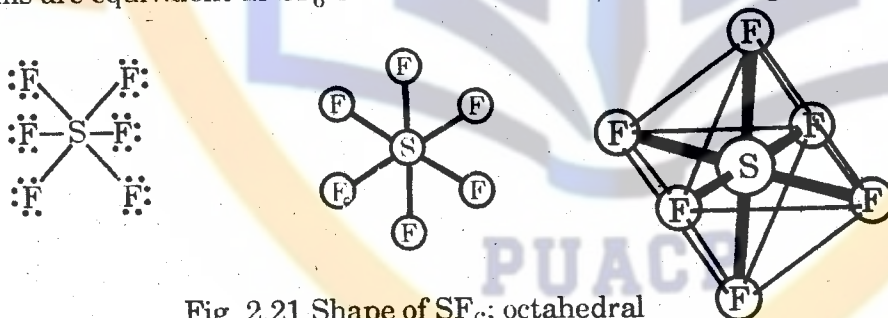
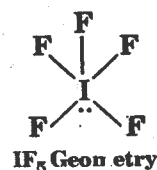


Fig. 2.21 Shape of SF₆; octahedral

The examples of such type molecules or ions are SF₆ and PCl_6^- etc.

AB₅E: Molecules in which the central atom is surrounded by six electron pairs, of which one is lone pair. The molecule adopts the shape of a square pyramid as shown in Table 2.2. Thus BrF₅, IF₅ and SbCl_5^{2-} etc., containing one lone pair would adopt square pyramidal geometry. In IF₅, the lone pair on iodine occupies one of the six equivalent positions in the octahedral arrangement, giving a square pyramidal geometry.



AB₄E₂: Molecules in which the central atom is surrounded by six electron pairs, out of which two are lone pairs and they are placed on opposite sides of the molecule, resulting in a square planar shape. Examples of this type are ICl_4^- , BrF_4^- etc., as shown in table 2.2. In XeF₄, the two lone pairs on Xe occupy opposing positions in the octahedral arrangement to minimize their repulsion. The result is a square planar geometry.

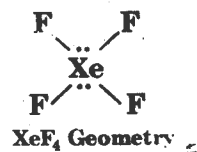
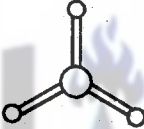
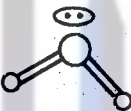
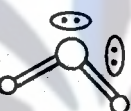


Table 2.2: Molecular geometries according to the VSEPR Theory

NO. of Electron Pairs	Number of lone pairs	Molecular geometry		Examples
2	0	Linear		$\text{BeCl}_2, \text{HgCl}_2$
3	0	Trigonal Planar		$\text{BF}_3, \text{BCl}_3, \text{HCHO}$
3	1	Trigonal		SO_2
4	0	Tetrahedral		$\text{CH}_4, \text{BF}_4^-$
4	1	Trigonal Pyramidal		$\text{NH}_3, \text{PCl}_3, \text{PH}_3, \text{AsCl}_3$
4	2	Bent (angular) V-shaped		$\text{H}_2\text{O}, \text{H}_2\text{S}$
5	0	Trigonal bipyramidal		$\text{PCl}_5, \text{SbCl}_5$
5	1	Seesaw (Irregular tetrahedral)		$\text{SF}_4, \text{TeCl}_4$
5	2	T-shaped		$\text{ClF}_3, \text{BrF}_3$
5	3	Linear		XeF_2

NO. of Electron Pairs	Number of lone pairs	Molecular geometry	Examples
6	0	Octahedral	 SF_6
6	1	Tetragonal Pyramidal (Square Pyramidal)	 $\text{BrF}_5, \text{IF}_5$
6	2	Square Planar	 ICl_4^-

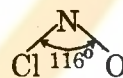
Example 2.1. Predict the geometry of the NOCl molecule.

Solution: i. The Lewis structure for NOCl is $\text{:}\ddot{\text{Cl}}-\ddot{\text{N}}=\ddot{\text{O}}\text{:}$

- The central atom is the N atom.
- There are two atoms and one lone pair on the N atom, so we count three valence shell electron pairs.
- VSEPR theory predicts trigonal planar geometry for three electron pairs.
- The atoms lie in the same plane but not on the same line.

A lone pair affects the structure more than a double bond.

Thus it has angular geometry and the measured angle is 116° .



2.12 Valence Bond Theory

The VSEPR theory is quite successful in explaining and predicting molecular geometry, but does not describe *how* bonding occurs and *where* bonding occurs, as well as where lone pairs of valence electrons are directed. Also, it does not permit a prediction of bond energies, bond lengths etc. In order to find the answer to these questions, the methods of wave mechanics have been employed. Two approaches have been used for the purpose of describing the covalent bond and electronic structures of molecules. Each approach employed quantum mechanics. The first approach, called the valence bond theory and the second called the molecular orbital theory. Both approaches are based on the concept of orbitals.

The valence Bond (VB) theory is based upon the assumption that half-filled atomic orbitals of combining atoms interact to form bigger new orbitals called *molecular orbitals*. These new orbitals are responsible for making the system to be a stable one. Heitler and London (1927) proposed that the electron wave of the valence orbital on one atom 'overlaps' the electron wave of the other bonding orbital to form a covalent bond. The essential conditions for the overlap of electron waves of orbitals are:

1. *The two overlapping orbitals must be valence orbitals, must be half-filled and must retain their identities.* All other orbitals on the atoms remain undisturbed.
2. *The half-filled valence orbitals of the atoms should approach sufficiently close to one another with the axis of their orbitals in proper alignment.*

Let us see how valence bond theory explains the formation of H_2 molecule starting from H atoms. We begin when two hydrogen atoms are far apart so that no interaction occurs. They are each described as either $1s_A$ or $1s_B$ and if electron 1 occupies atom A and electron 2 occupies atom B, the wave function for the pair is:

$$\Psi = \psi_{1s_A}(1) \psi_{1s_B}(2) \quad (1)$$

Note that the wave functions of separate systems are multiplied to produce the overall function. This is an accurate description when the atoms are infinitely separated.

As the two hydrogen atoms approach each other, the electron of one atom is attracted to the nucleus of the other. The electrostatic attraction increases steadily as the distance shortens. However, repulsion between electron-electron and nucleus-nucleus also become important. The attractive and repulsive forces are balanced at an internuclear distance of 0.074 nm, the actual distance between two hydrogen nuclei in an H_2 molecule and the system has its lowest (minimum) energy-

436 kJ/mol corresponding to the most stable arrangement. The plot of potential energy versus internuclear distance for two hydrogen atoms is shown in Fig. 2.22. The valence bond theory assumes that it is also an adequate description when the atoms are at their equilibrium separation in the molecule. In the electrostatic analysis, we treat the electrons as distinguishable particles. It turns out that we really do not know which electron is which i.e., it is impossible to say whether it is electron 1 that is in $1s_A$ and electron 2 in $1s_B$, or vice versa. When we take this into account, the electrons are given a little more freedom, which leads to a lower energy. In quantum mechanics, this effect is described in terms of

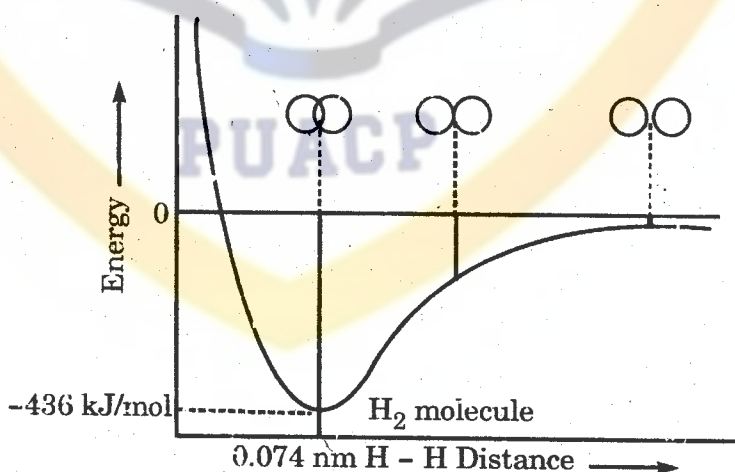


Fig. 2.22. Energy of the H_2 molecule as a function of the distance between the two nuclei. The lowest point on the curve -436 kJ/mol, corresponds to the internuclear distance observed in the H_2 molecule, 0.074 nm. Energy is compared to that of two separated hydrogen atoms, which is defined as zero.

"orbital overlap". We say that H_2 molecule is stabilized by the overlap of the two $1s$ orbitals. Each electron, in effect, is spread over both orbitals. Therefore, the wave function $\psi_{1s_A}(2)$ and $\psi_{1s_B}(1)$ is admitted as an equally valid description and the valence bond wave function $\psi_{v.b.}$ for the pair of electrons in the molecule may be taken to be a linear combination of the two products and is written as:

$$\psi_{v.b.} = \psi_{1s_A}(1) \psi_{1s_B}(2) + \psi_{1s_A}(2) \psi_{1s_B}(1) \quad (2)$$

In H_2 molecule, because of symmetry, the wave function would contribute equally well. The probability density of the electrons in the molecule is proportional to the square of the VB wave function:

$$\psi_{v.b.}^2 = \psi_{1s_A}(1)^2 \psi_{1s_B}(2)^2 + \psi_{1s_A}(2)^2 \psi_{1s_B}(1)^2 + 2 \psi_{1s_A}(1) \psi_{1s_B}(1) \psi_{1s_A}(2) \psi_{1s_B}(2) \quad (3)$$

Such an interaction is more like a covalent bond in which one electron is shared equally between the two atoms. The terms $\psi_{1s_A}(1) \psi_{1s_B}(2)$ and $\psi_{1s_A}(2) \psi_{1s_B}(1)$ represent enhancement of electron density in the overlap region. Therefore, the strength of the bond in VB theory can be a measure of accumulation of electron density in the internuclear region. The VB approach treats the bond as purely covalent.

The valence bond description of H_2 (and any other diatomic molecule) could be improved by adding terms that permit both electrons to appear in the same atom to some extent. A wave function that puts both electrons into $1s_A$ is $\psi_{1s_A}(1) \psi_{1s_A}(2)$. If both are in $1s_B$ the wave function contains a term $\psi_{1s_B}(1) \psi_{1s_B}(2)$. Since both H^-H^+ and H^+H^- are equally probable the original valence bond wave function ought to be augmented by an ionic contribution;

$$\psi_{ionic} = \psi_{1s_A}(1) \psi_{1s_A}(2) + \psi_{1s_B}(1) \psi_{1s_B}(2) \quad (4)$$

The new, improved wave function is therefore ;

$$\psi = \psi_{v.b.(\text{Covalent})} + c \psi_{ionic}$$

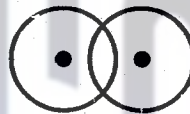
Where the value of c governs the extent of ionic character mixed into the original purely covalent v.b. structure. The improvement of the wave function by this admixture is called **ionic-covalent resonance**, and the improvement in energy can be traced to the extra freedom it gives to the distribution of the electrons.

The actual value of c is about 0.25, which indicates that the ionic structures contribute about $c^2 \sim 0.06$ or 6 per cent to the overall structure of H_2 . In heteronuclear molecules the ionic terms are much more important, and in species like $NaCl$ they wholly dominate the vestigial covalent character.

The formation of an electron pair covalent bond in H_2 molecule is based on the following assumptions.

1. Each hydrogen atom has a partially filled $1s$ atomic orbital, each having one electron.

- Heitler and London pointed out that the electrons are indistinguishable and the wave function must express this fact.
- The molecular wave function $\psi_{v.b.}$ is obtained by the linear combination of the wave functions of the atomic orbitals using the concept of resonance.
- The pairing of electrons in the molecule should satisfy the Pauli's exclusion principle and the two electrons should have different values of spin quantum numbers.
- An attractive force (bond energy) is the result of electron pairing.
- The electron pair is localized between the two nuclei.
- The overlapping orbitals have similar symmetry and energy. The orbitals overlap each other to an appreciable extent. Greater the overlap stronger will be the bond formed.



Overlap of 1s orbitals in H_2
(σ bond)

The description of the H_2 molecule may be generalized that a *covalent bond arises from the exchange forces between a pair of electrons with opposite spin.*

2.13 Sigma (σ) and Pi (π) Bonds

A molecular orbital (MO) which is symmetrical about the line joining the two nuclei and is occupied by a pair of electrons, is called a **sigma (σ) molecular orbital (σ -MO)** and the linkage thus established is called a **sigma bond**. Although the electrons have freedom within the entire region of MO, the probability of finding these electrons is greatest between the two nuclei on the bond axis. A straight line joining the two nuclei of bonded atoms is called **bond axis**. All single bonds are sigma bonds and they are formed by end-to-end or head on overlap of the atomic orbitals. A single bond is formed by the overlap of two s-orbitals, an s-orbital with one p-orbital, or two p orbitals as shown in Fig. 2.23. A bond in which the electron distribution is concentrated along the internuclear axis and possesses axial symmetry is called **sigma (σ) bond**.

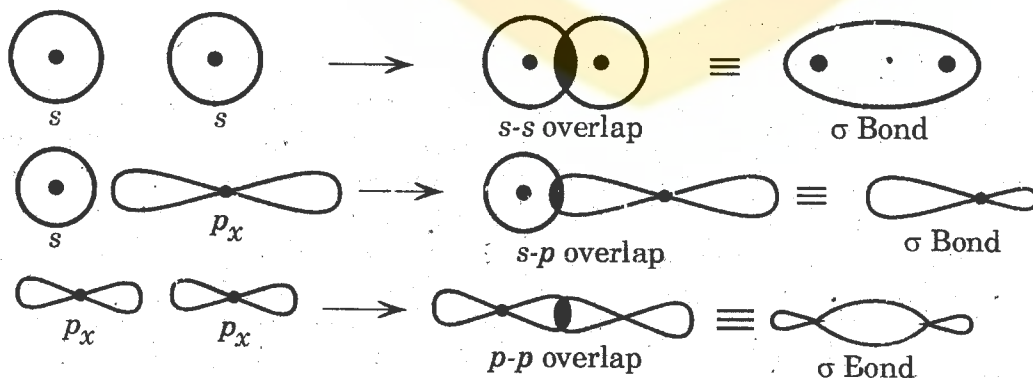
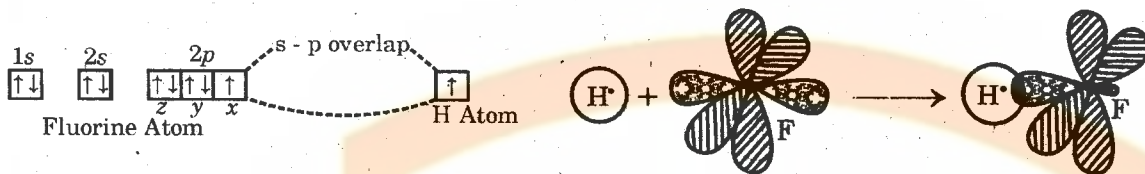


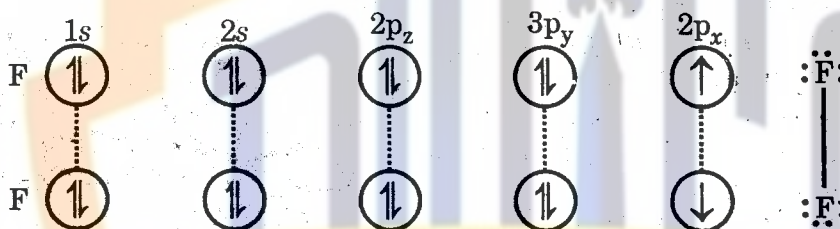
Fig. 2.23. Formation of σ bonds

In the formation of **hydrogen fluoride** molecule, the half-filled 1s orbital of hydrogen overlaps with the half-filled $2p_x$ orbital of F atom and this allows pairing of

electrons to form σ bond. The electronic configuration of F atom is $1s^2 2s^2 2p_x^1 2p_y^2 2p_z^2$.



The formation of **Fluorine** molecule can be represented as:



Pi (π) Bond: A bond resulting from the side-wise overlap of two half-filled, parallel p -orbitals in which the regions of electron density (cloud) are above and below the internuclear axis is called π bond. The internuclear axis lies in the nodal plane. The molecular orbital (MO) thus formed is called Pi (π) molecular orbital. It should be noted that the p orbitals have parallel axes and should be coplanar (in one plane). It should be noted that:

- A π molecular orbital involves an overlap at both the lobes of the p -orbitals, whereas in a sigma bond the overlap takes place in a single region.
- Sigma (σ) molecular orbital do not have a nodal plane which contains the nuclear axis, whereas a π MO possesses a nodal plane containing the nuclei and dividing the orbital into two halves. π MO possesses 'sausage-like' regions parallel to the nodal plane passing through the two nuclei.

Let us take the example of oxygen molecule. Oxygen atom has the electronic configuration $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^2$. Each oxygen atom has two half-filled p orbitals. The two p_x orbitals overlap end to end to form a σ bond. The $2p_y$ orbitals overlap sidewise to form a π bond. The formation of σ and π bonds in O_2 molecule is shown in Fig. 2.24.

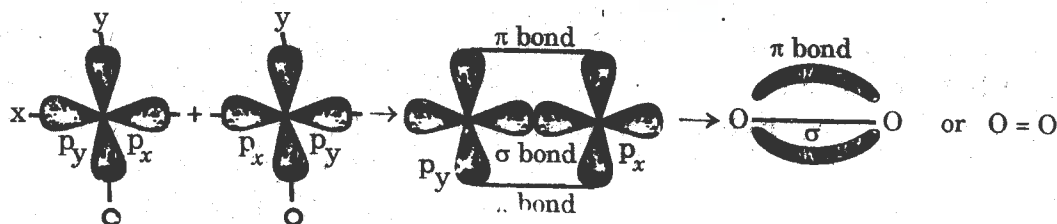
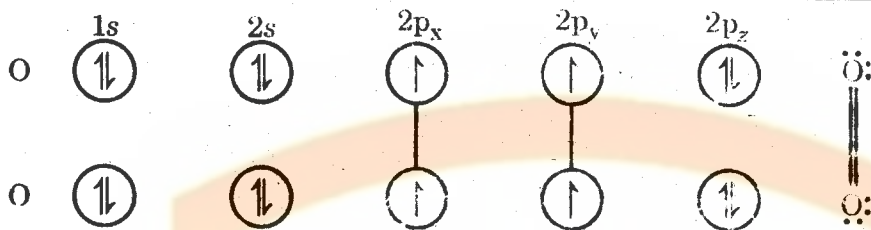


Fig. 2.24. σ and π -bonds formation in O_2 molecule

The formation of O_2 molecule based on VB theory can be represented as:



According to valence bond theory, O_2 molecule should have a double bond with no unpaired electrons. But the paramagnetic nature and magnetic moment value of O_2 indicates the presence of two unpaired electrons. This is an example of the failure of the VB theory in predicting the electronic structure.

As already pointed out, the π MO formed here is a pair of 'sausage-like' regions lying parallel to the σ bond. The π bond is analogous to the classical **double bond** of organic chemistry. The formation of N_2 molecule is shown in Fig. 2.25.

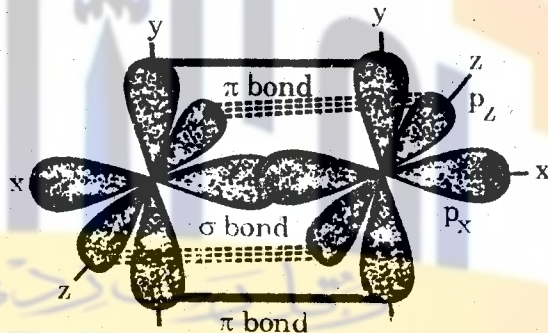


Fig. 2.25 Formation of one σ bond and two π bonds in N_2 molecule.

We conclude that every simple double bond consists of a sigma (σ) and a pi (π) bond and every triple bond contains one σ -bond and two π -bonds.

2.14 Hybridization of Orbitals

Many elements, such as Be, B and C, form compounds whose formation cannot be explained on the basis of ground-state electronic configuration. Beryllium whose electronic configuration is $1s^2 2s^2$ would be inert and forms no bond at all. In fact, Be is not an inert element as it reacts with other elements to form compounds showing its bivalency e.g. BeH_2 , $BeCl_2$. Boron which has the electronic configuration $1s^2 2s^2 2p_x^1$, should behave as a monovalent element, but actually B forms trivalent compounds such as BF_3 , BCl_3 . Carbon with electronic configuration $1s^2 2s^2 2p_x^1 2p_y^1$ has two unpaired electrons and it should be divalent, but it is found to be tetravalent in its compounds such as CH_4 , CCl_4 .

Pauling and Slater resolved this discrepancy by introducing the concept of **orbital hybridization** which involves mixing of different orbitals (say s and p) of an atom to form new **hybrid orbitals**, before overlapping and bond formation. These hybrid orbitals are identical in directional character and have the same shape and energy. *The total number of hybrid orbitals formed after mixing, is always equal to the number of atomic orbitals mixed.* The hybrid orbitals are just atomic orbitals, although they arise only during the process of bonding and do not exist in the free atom. A bonding process can then be considered as a process of overlapping the

hybrid orbitals of one atom with a pure or hybrid atomic orbital of another.

"The process of mixing of orbitals of an atom of nearly equal energy giving rise to entirely new orbitals equal to the number of mixing orbitals and having identical shape and same energy content is called hybridization".

The hybrid orbitals are named by indicating the *number* and *kind* of atomic orbitals hybridized. For example, hybridization of *one* 2s orbital and *one* 2p orbital gives *two* sp hybrid orbitals.

Types of the hybridization are:

- (i) sp (ii) sp^2 (iii) sp^3 (iv) dsp^2 (v) dsp^3 (vi) sp^3d^2 or d^2sp^3 (vii) sp^3d^3

(i) sp hybridization (the formation of $BeCl_2$)

The divalency of Be can be explained by assuming that one of the 2s electrons is promoted to a vacant 2p orbital to give the configuration $1s^2 2s^1 2p_x^1$. The Be atom now has two half-filled orbitals which can form two covalent bonds by overlapping with the half-filled $3p_x$ orbitals of two chlorine atoms. The two Be-Cl bonds thus formed would be *nonequivalent*, because the Be 2s and 2p orbitals are not expected to overlap Cl $3p_x$ orbital with equal effectiveness. This is contrary to the fact that both Be-Cl bonds are equivalent in bond length and bond strength. Thus Be atom does not use simple 2s and 2p orbitals individually, but 2s and 2p orbitals are mixed to form two new equivalent orbitals called **sp hybrid orbitals**. These sp orbitals are oriented at an angle of 180° . The sp hybrid orbital has two lobes, one with greater extension in space than the other.

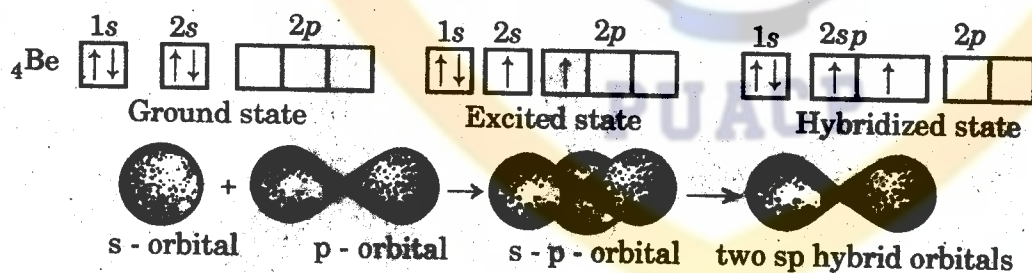


Fig. 2.26. Formation of sp hybrid orbitals

The formation of $BeCl_2$ can be visualized as the overlap of the $2p_x$ orbitals of chlorine atoms with the two equivalent collinear sp hybrid orbitals of Be atom. The two Cl - Be - Cl bonds must be equivalent and collinear.



Fig. 2.27. Bond formation in $BeCl_2$ using sp hybrid orbitals

In general, sp hybridization occurs at the central atom of a molecule or a polyatomic ion, whenever there are two regions of high electron density around the central atom. This kind of hybridization is called **digonal hybridization**. The

structures of BeF_2 , BeBr_2 and BeI_2 are similar to that of BeCl_2 . The chlorides, bromides and iodides of Cd and Hg are also linear covalent molecules.

(ii) sp^2 Hybridization (BF_3 molecule)

The mixing of one s orbital and two p orbitals on the same atom to form three new equivalent hybrid orbitals is called sp^2 hybridization. These three sp^2 hybrid orbitals lie in the same plane and are directed towards the three corners of an equilateral triangle forming a mutual angle of 120° . It is for this reason that sp^2 orbitals are also called **trigonal hybrid**, the process being referred to as **trigonal hybridization**.

The electronic configuration of boron is $1s^2, 2s^2 2p^1$. It has only one unpaired electron, which demands the formation of only one covalent bond. Trivalency of boron can be explained on the basis of sp^2 hybridization. Boron must, therefore, exist in the excited state when one s electron is promoted to p orbital to give the configuration $1s^2, 2s^1, 2p_x^1 2p_y^1$. Now in the excited boron, we have three unpaired orbitals, so it can form three covalent bonds. However, not only would we predict that two of the bonds would be different from the third (because two $2p$ orbitals on B and one $2s$ orbital on B would be involved), but also we could not account for the observed 120° bond angles. So the evidence in favour of hybridization is very strong. Therefore, boron utilizes its $2s$, $2p_x$ and $2p_y$ orbitals for the formation of $3sp^2$ hybrid orbitals. These three hybrid orbitals overlap with three p orbitals from three different F atoms to form three B-F bonds at an angle of 120° with each other.

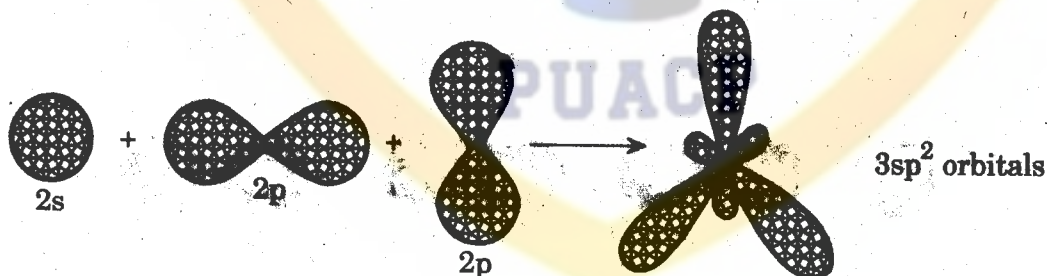


Fig. 2.28 Formation of three sp^2 hybrid orbitals

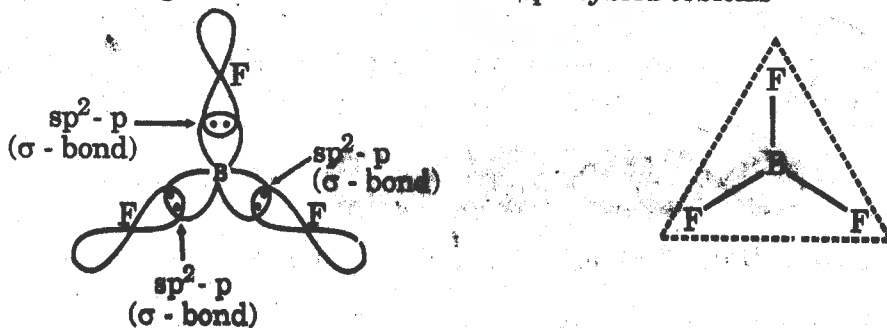


Fig. 2.29. Formation of triangular planar BF_3 molecule

In general, sp^2 hybridization occurs at the central atom whenever there are three regions of high electron density around the central atom.

(iii) sp^3 Hybridization (methane, CH_4)

The mixing of one s and three p orbitals on a single atom to form four equivalent sp^3 hybrid orbitals is called sp^3 hybridization. The tetravalency of carbon can be explained on the basis of sp^3 hybridization.

Carbon with an electronic configuration $1s^2 2s^2 2p_x^1 2p_y^1$ has two unpaired electrons and would form two covalent bonds, but forms four bonds in most of its compounds. This has been explained that one of the two s electrons is promoted to an empty $2p$ orbital giving the excited state a configuration of carbon $1s^2 2s^1 2p_x^1 2p_y^1 2p_z^1$. In the excited state carbon can form four bonds but these would not be equivalent. In actual practice all the four C-H bonds in CH_4 are equivalent with a bond angle of 109.5° . This has been explained by the process of hybridization in which $2s$ and three $2p$ orbitals combine to form four equivalent sp^3 hybrid orbitals. Each sp^3 orbital is composed of s and p orbitals in the ratio of 1:3 and is directed toward the corners of a regular tetrahedron, forming an angle of 109.5° to each other. This type of hybridization is called **tetrahedral hybridization**.

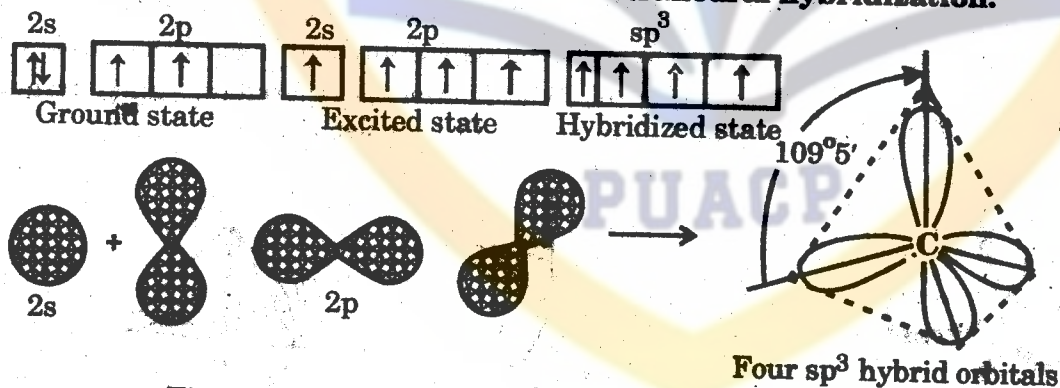


Fig. 2.30. Formation of $4sp^3$ hybrid orbitals.

The methane molecule is formed by the overlap of four carbon sp^3 orbitals with $1s$ orbitals of four hydrogen atoms. The structure of CH_4 is considered tetrahedral.

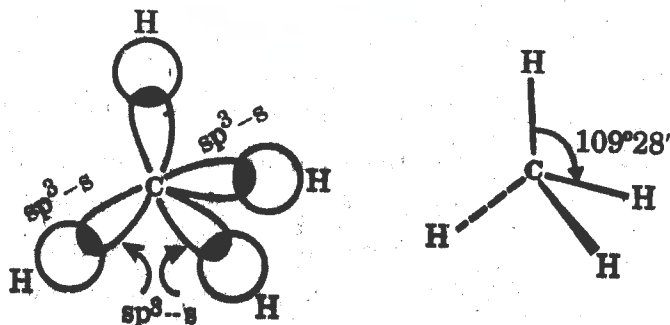


Fig. 2.31 Tetrahedral shape of CH_4 molecule

Hybridization involving *d* orbitals

There are several types of hybridization involving *d* orbitals. Since the *d* orbitals have a relatively complex shape, we will consider here only some of the common types. The most important of these are dsp^2 hybridization, dsp^3 hybridization and d^2sp^3 hybridization.

(iv) **dsp^2 hybridization**, would occur when one *s* orbital, two *p* orbitals (p_x, p_y) and one *d* orbital (from the next to outermost shell, $d_{x^2-y^2}$) mix together to give four equivalent

hybrid orbitals. These four hybrid orbitals are directed to the corners of a square in the *xy* plane. This gives a square planar arrangement between them and the hybridization is, therefore, called **square planar hybridization**. Many compounds of nickel and platinum possess this square planar shape e.g., $[\text{Pt}(\text{NH}_3)_4]^{2+}$, $\text{Ni}(\text{CN})_4^{2-}$.

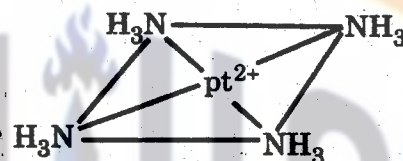
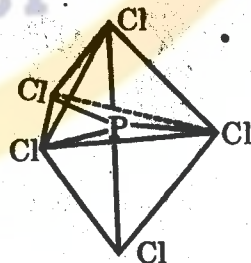
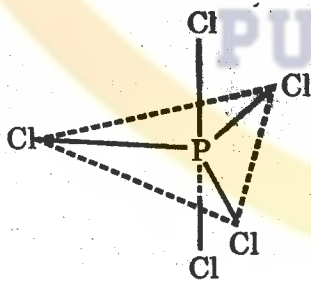


Fig. 2.32 Square planar structure of $[\text{Pt}(\text{NH}_3)_4]^{2+}$

(v) **dsp^3 hybridization** would occur when one *s* orbital, 3*p* orbitals and one *d* (d_{z^2}) of the outermost electronic shell, mix together to give five hybrid orbitals. These hybrid orbitals so formed are not equivalent and consist of one group of two equivalent oppositely directed orbitals and a second group of three equivalent orbitals. The five sp^3d or dsp^3 hybrid orbitals are directed toward the corners of a trigonal bipyramid and this type of hybridization is, therefore, called **trigonal bipyramidal hybridization**. An example is PCl_5 molecule.



sp^3d orbitals

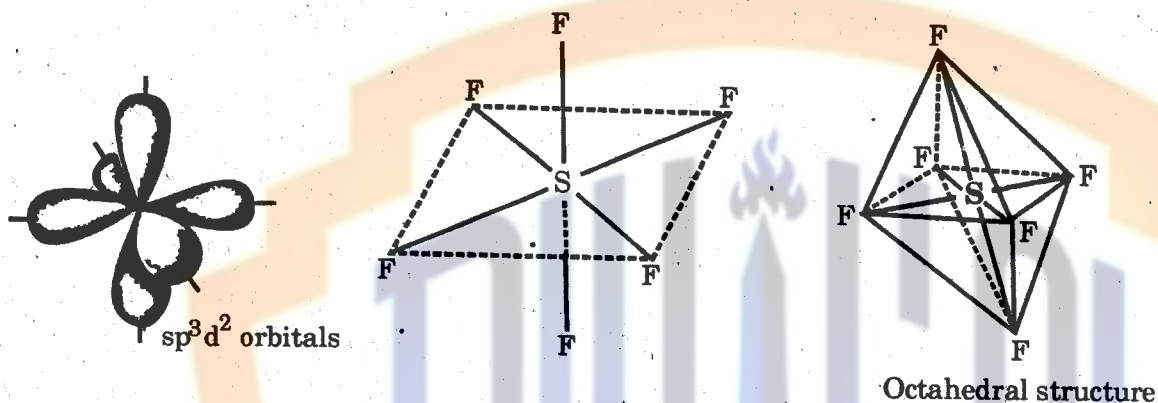


Trigonal bipyramidal structure

Fig. 2.33. dsp^3 bonding configuration

d^2sp^3 or sp^3d^2 hybridization occurs when one *s*, three *p* and two *d* ($d_{x^2-y^2}$ and d_{z^2}) mix together to form six equivalent hybrid orbitals. The six sp^3d^2 hybrid orbitals are directed toward the corners of a regular octahedron. Four out of six hybrid orbitals are lying in one plane and are inclined to one another at 90° while the remaining two are directed above and below the plane containing the four hybrid orbitals perpendicularly. The formation of SF_6 molecule is an example of

sp^3d^2 hybridization. Each sp^3d^2 hybrid orbital is overlapped by a half-filled p orbital from six fluorine atoms to form a total of six covalent bonds.



(vii) sp^3d^3 Hybridization ($s + p_x + p_y + p_z + d_{xy} + d_{yz} + d_{zx}$ AO's). Mixing of one s three p and three d atomic orbitals gives seven sp^3d^3 hybrid orbitals which are directed towards the corners of a pentagonal-bipyramid. These are not equivalent hybrid orbitals. Five of them point towards the vertices of a regular pentagon while the other two are oriented perpendicular (90°) to the plane containing the first set of five hybrids. Examples are IF_7 molecule and XeF_6 molecule.

IF_7 molecule. The valence electronic configuration of iodine atom (central atom) in the ground state is $5s^2, 5p_x^2, 5p_y^2, 5p_z^1$. In order to make available seven singly-occupied orbitals for sp^3d^3 hybridization one $5s$ and two $5p$ electrons are promoted to the vacant high energy $5d_{xy}$, $5d_{yz}$ and $5d_{zx}$ orbitals. These seven atomic orbitals then hybridize to form seven sp^3d^3 hybrid orbitals all of which are singly-filled and, therefore overlap with seven singly-occupied $2p_z$ atomic orbitals of seven fluorine atoms to form seven I-F σ bonds. All these are $sp^3d^3-p_z$ σ bonds and the molecule is pentagonal bipyramidal in shape (see Fig).

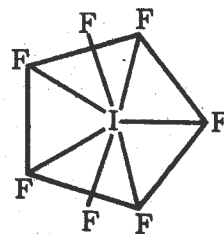
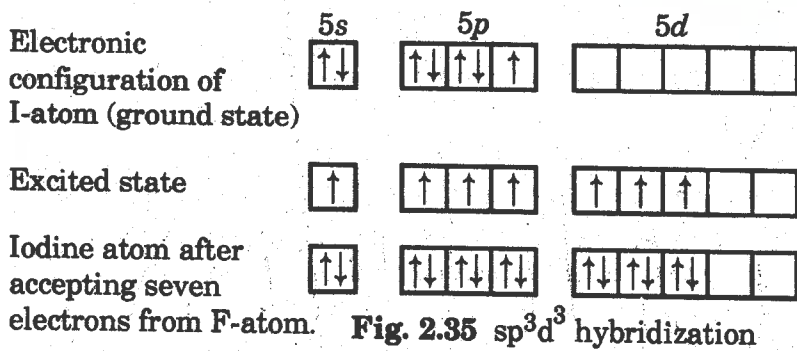


Fig. 2.36 Pentagonal bipyramidal IF_7 molecule.

Axial I-F distances are not the same as equatorial I-F distances. This confirms the nonequivalent nature of sp^3d^3 hybrid orbitals.

Table 2.3. Hybrid orbitals and their geometrical orientation

No. of electron pairs in outer shell	Hybrid orbitals	Orientation	Bond Angle	Examples
2	sp	Linear	180°	$BeCl_2, C_2H_2$
3	sp^2	Trigonal	120°	BF_3, GaI_3
4	sp^3	Tetrahedral	109.5°	CH_4, CCl_4, SiF_4
4	dsp^2	Square planar	90°	$Pt(NH_3)_4^{2+}, Ni(CN)_4^{2-}$
5	dsp^3	Trigonal planar	120° & 90°	PCl_5
6	d^2sp^3	Octahedral	90°	$SF_6, [Co(NH_3)_6]^{2+}$
7	sp^3d^3	Pentagonal bipyramidal		IF_7, XeF_6

Summary of bridization

The important steps in hybridization are summarized as follows:

1. The formation of an excited state which involves unpairing of electrons followed by promotion of electrons to orbitals of higher energy. The promotion of electrons may take place between orbitals with the same principal quantum number n or between orbitals with different n values.
2. The pure atomic orbitals in the excited state are mixed or hybridized to form equivalent hybrid orbitals.
3. These hybrid orbitals have a definite orientations in space.

2.15 Molecular Orbital Theory (MOT)

The valence bond theory is based upon the assumption that the formation of a molecule involves an interaction between the electron waves of only the atomic orbitals of participating atoms which are half-filled. These bonding orbitals (half-filled) merge into one another to give a new orbital of a bigger size which is responsible for the stability of the system. All other orbitals on the atoms remain undisturbed.

According to the molecular orbital theory (developed by Hund, Mulliken and Huckel), all the valence electrons in molecule are associated with all the nuclei concerned. In other words, all the valence electrons have an influence on the stability of the molecule. (Inner-shell electrons may also make a contribution to the bonding but for many simple molecules the effect is small). Further more, MO theory considers that valence-shell atomic orbital cease to exist when a molecule is formed. They are replaced by a new set of energy levels with corresponding new charge-cloud distributions. These new energy levels are a property of the molecules as a whole, and are called, consequently, *molecular orbitals*. "An electronic energy level in a molecule and the corresponding charge-cloud distribution in space is called

molecular orbital". Molecular orbitals may be obtained by the linear combination of atomic orbitals (LC AO method) belonging to each atom in the molecule. The wave functions of the atomic orbitals are combined mathematically to produce wave functions for the resulting molecular orbitals. The molecular orbitals are polycentric and not monocentric as in the case of an atom. The number of molecular orbitals produced is equal to the number of original atomic orbitals combined. The rules for filling the electrons in these molecular orbitals are the same as for filling the atomic orbitals.

Let us consider the formation of a simple homonuclear diatomic molecule such as hydrogen molecule in which two identical atoms are linked by an electron pair. Although the atoms are identical but it will be convenient to distinguish the two atoms by writing, H_A and H_B . Each hydrogen atom has a single electron in $1s$ orbital. Let the wave functions for describing the two $1s$ atomic orbitals be ψ_A and ψ_B for hydrogen atoms H_A and H_B . The effective overlap of the wave functions ψ_A and ψ_B will take place only if (i) the orbitals have similar energy state (ii) the orbitals overlap to a considerable extent, and (iii) orbitals have the same symmetry. All these conditions are fulfilled by atomic orbitals of both the hydrogen atoms. The molecular orbital wave function ψ will be obtained by the linear combination of the atomic orbital wave function ψ_A and ψ_B .

$$\psi = C_A \psi_A (1s) \pm C_B \psi_B (1s) \quad (6)$$

where C_A and C_B are mixing coefficients which may be replaced by single coefficient, λ . Thus

$$\psi = \psi_A (1s) \pm \lambda \psi_B (1s) \quad (7)$$

λ gives the relative proportions of $\psi_A (1s)$ and $\psi_B (1s)$ in the molecular orbital in the ratio of $1^2 : \lambda^2$. Since both H_A and H_B are identical, thus atomic orbitals must make equal contributions to the molecular orbitals so that $\lambda^2 = 1$ or $\lambda = \pm 1$. Thus by substituting the values of λ in equation (7), two possible molecular orbitals wave functions are obtained, one from the addition and other from the subtraction i.e.,

$$\psi_{\text{bonding}} = \psi_A (1s) + \psi_B (1s) \quad (8)$$

$$\psi_{\text{antibonding}} = \psi_A (1s) - \psi_B (1s) \quad (9)$$

Since the probability of finding an electron in a particular space is expressed by, ψ^2 , squaring of equations (8) and (9) will give the expression for the probability of finding an electron at any point within the molecular orbitals and from this expression the corresponding boundary surfaces and also energy levels can be found.

$$\psi_b^2 = \psi_A^2 (1s)^2 + \psi_B^2 (1s)^2 + 2\psi_A (1s) \psi_B (1s) \quad (10)$$

$$\psi_a^2 = \psi_A^2 (1s)^2 + \psi_B^2 (1s)^2 - 2\psi_A (1s) \psi_B (1s) \quad (11)$$

The boundary surfaces and the relative energies of two MOs which are formed by combining two $1s$ AOs are shown in Fig. 2.37.

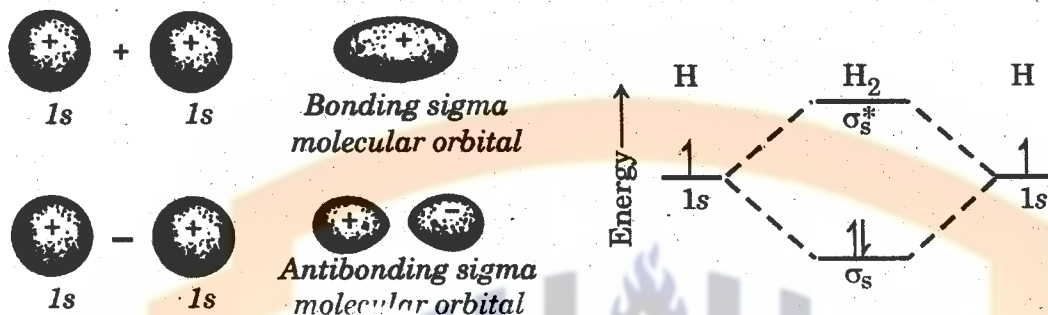


Fig. 2.37. (a) Combination of 1s atomic orbitals to form MOs. (b) Relative energy of σ_{1s} orbitals in H_2 molecule.

ψ_b molecular orbital is formed by the addition of the two 1s orbitals when atomic orbitals of the same sign overlap along the internuclear axes. Such overlap leads to reinforcement of the wave functions in the region between the two nuclei, and ψ_b the probability of finding the electron is large between the nuclei, resulting a strong bond between the atoms. The bonding molecular orbital (ψ_b) is in a lower energy state than the average energies of the combining atomic orbitals. The bonding molecular orbitals of the type which are formed from 1s atomic orbitals are denoted as $\sigma(1s)$. ψ_a is formed by the subtraction of the two 1s orbitals, when orbitals of opposite sign overlap, the wave functions interfere with each other in the region between the two nuclei and a node is produced. At the node $\Psi = 0$, and on either side of the node ψ is small, therefore, ψ_a^2 is also small in the region between the nuclei. The repulsive forces will be greater than the attractive forces and therefore, antibonding molecular orbital is at a state of higher energy than the average energies of the component atomic orbitals. The antibonding molecular orbital of this type is denoted by $\sigma^*(1s)$. Both bonding and antibonding molecular orbitals have cylindrical symmetry about the internuclear axis, therefore these orbitals are called σ orbitals and the bonds are called σ bonds.

Conditions for effective combination of atomic orbitals

There are certain conditions for effective combination of atomic orbitals which are described below:

- (i) The energies of the AO's combining together must be similar in magnitude or the AO's should have comparable energies. Thus in case of the formation of a homonuclear diatomic molecule of A_2 type, the $1s_a$ -AO of the atom A_a will not combine with the $2s_b$ -AO of another atom A_b of the same element, where A_a and A_b are the two atoms of the molecule A_2 since their energies are not equal. Similarly since the energy difference between 2s- and 2p-AO's is too great, they will also not combine. But in case of the formation of heteronuclear diatomic molecule of AB type, such combination may be expected.

(ii) The charge clouds of the AO's must overlap one another as much as possible, if they are going to combine together to form the MO's. This condition is referred to as the *principle of maximum overlap*.

(iii) The AO's should have the same symmetry about the molecular axis. This condition is known as *symmetry condition for the combination of AO's*.

On the basis of this symmetry condition it is noted that some of the AO's which have comparable energies do overlap, but cannot combine to give MO's. Thus MO's cannot be formed by the overlap of an *s*-atomic orbital of an atom A and one *p*-atomic orbital of atom B perpendicular to the molecular axis (either p_x - or p_y -orbital), since the molecular axis is the *z*-axis. The cause of non-formation of MO's is that the symmetry of *s*-orbital is not the same as that of *p*-orbital. Alternatively it can be said that $++$ overlap is neutralized by the $+ -$ overlap or in other words the $+ -$ overlap cancels the bonding contribution from the $++$ overlap.

Thus the following pairs of AO's will not combine to form any MO's, provided that *z*-axis is assumed to be the molecular axis:

(a) $s - p_x$ pair (b) $s - p_y$ pair (c) $p_x - p_y$ pair (d) $p_x - p_z$ pair (e) $p_y - p_z$ pair.

H₂ molecule has two electrons, both in bonding σ_s orbital and constitute a single covalent bond.

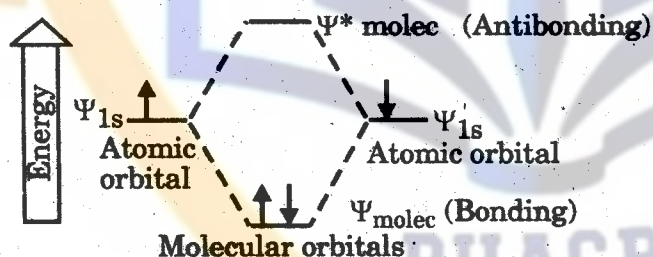


Fig 2.38. Molecular orbital diagram for H₂ molecule

In molecular orbital theory, the **bond order** is defined as one-half the difference between the numbers of electrons in bonding and in antibonding molecular orbitals (MOs) in a molecule.

$$\text{Bond order} = \frac{\text{No. of electrons in bonding MO} - \text{No. of electrons in antibonding MO}}{2}$$

$$\text{The bond order in H}_2 \text{ molecule} = \frac{2 - 0}{2} = 1$$

Bond order indicates whether a bond is single, double, triple (or one-half, three halves, five halves); whereas Lewis theory and Valence bond theory require electron-pair bonds, MO theory accounts quite nicely for one-electron bonds which are known to exist.

Formation of MOs for He₂,

Consider the He₂ molecule which might be formed from two He atoms, each

of which furnishes two electrons to the molecule. Thus He_2 molecule will have four electrons, two in the σ_{1s} bonding MO and two in the σ_{1s}^* antibonding MO. As a result of this, the stability gained by two electrons moving to low energy σ_{1s} bonding MO would be lost by the other two electrons moving to the σ_{1s}^* antibonding MO. The result is that no attractive force between He atoms and so He_2 does not exist.

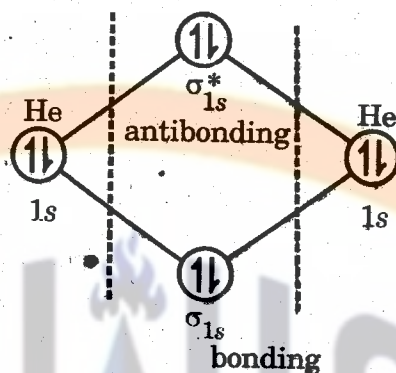


Fig. 2.39. Diagram for the hypothetical for He_2 molecule

$$\text{Bond order in } \text{He}_2 = \frac{2-2}{2} = 0$$

Second period elements; Homonuclear diatomic molecules

In the elements of the second period the $1s$ -orbitals are sucked very close to the nuclei, and being so small they overlap to only a small extent, and contribute very little to the bonding energy. The $2s$ -orbitals extend over a significant region, and overlap significantly. Therefore, the molecular orbitals they form play a significant role in the energies of the second period diatomics. We have seen that two $1s$ -orbitals can be combined to form two MO: one bonding and one antibonding. The same is true of two $2s$ -orbitals.

In the second period diatomics we also have to allow for the overlap of the $2p$ -orbitals. The combination of two p orbitals produces different results depending on which p -orbitals are used. If the x -axis is the internuclear axis, then two $2p_x$ orbitals can overlap properly if they approach each other end to end, to form two MOs: one σ_{2p} bonding orbital with electronic charge built up between the nuclei, and the second σ_{2p}^* antibonding MO with decreased charge density between the nuclei. The bonding and antibonding MOs can be described in terms of wave functions ψ_b and ψ_a , respectively, as follows:

$$\psi_b = \psi_A(2p_x) + \psi_b(2p_x) \text{ termed as } \sigma_{2p_x}$$

$$\psi_a = \psi_A(2p_x) - \psi_b(2p_x) \text{ termed as } \sigma_{2p_x}^*$$

Thus ψ_b is a bonding MO and ψ_a is an antibonding MO.

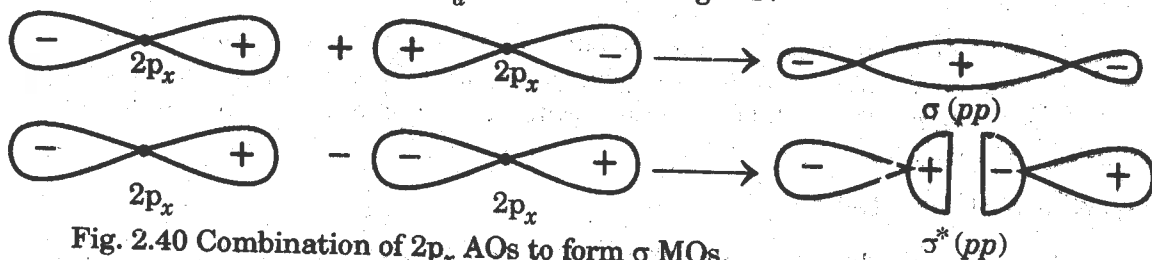


Fig. 2.40 Combination of $2p_x$ AOs to form σ MOs.

When $2p_y$ and $2p_z$ orbitals overlap, they do so in a parallel or sidewise fashion to form MOs which are not symmetrical about the line joining the nuclei. These unsymmetrical combinations are called π orbitals for bonding and π^* orbitals for antibonding, but both are weaker than in the σ case because the accumulation of electron density lies away from the optimum positions. There are two MOs of π type and two MOs of π^* because there are two pairs of p orbitals that are arranged in a parallel fashion. These MOs obtained from p_y and p_z AOs can be described in terms of wave functions ψ_b and ψ_a as follows:

- (i) Combination of two p_y atomic orbitals:

$$\psi_b = \psi_A(2p_y) + \psi_B(2p_y) \text{ termed as } \pi_{2p_y} \quad (\text{bonding})$$

$$\psi_a = \psi_A(2p_y) - \psi_B(2p_y) \text{ termed as } \pi_{2p_y}^* \quad (\text{antibonding})$$

- (ii) Combination of two p_z atomic orbitals

$$\psi_b = \psi_A(2p_z) + \psi_B(2p_z) \text{ termed as } \pi_{2p_z} \quad (\text{bonding})$$

$$\psi_a = \psi_A(2p_z) - \psi_B(2p_z) \text{ termed as } \pi_{2p_z}^* \quad (\text{antibonding})$$

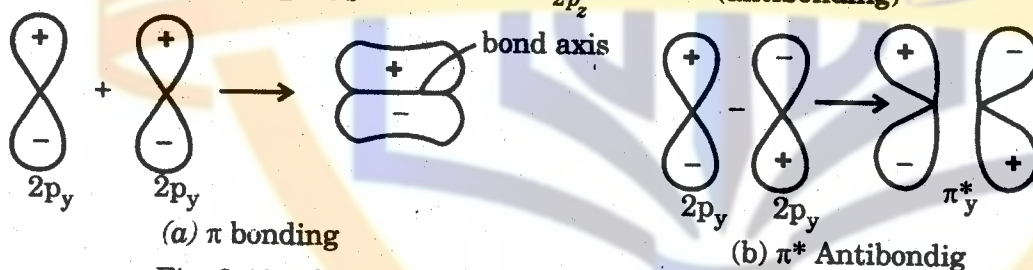


Fig. 2.41. Combination of 2p AOs to form π Mos π_{2p_y}

In order to know about the filling of electrons in MOs it is necessary to find out the order of energy levels or stability of their MOs.

The order of increasing energy levels are revealed by spectroscopic data for B_2 through N_2 , is as follows:

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2p_y} = \pi_{2p_z} < \sigma_{2p_x} < \pi_{2p_y}^* = \pi_{2p_z}^* < \sigma_{2p_x}^*$$

However the relative energy of p_y and p_z orbitals is higher than that of the σ_{2p_x} orbital for O_2 , F_2 and (hypothetical) Ne_2 .

The change in sequence of MO energies between N_2 and O_2 occurs because the σ_x and σ_x^* MOs actually have some s character, a fact which we had to ignore, when we decided to use the "one AO plus one AO yields two MOs" simplification. The amount of s character in these orbitals decreases as the nuclear charge increases across the period. Because of this the σ_x energy drops below the $\pi_y - \pi_z$ energy at O_2 .

The π_{2p_y} and π_{2p_z} bonding and antibonding MOs have the same energy, i.e; they are degenerate.

Li₂ : The electronic configuration of each Li atom (At. No. 3) is $1s^2 2s^1$. The first shell (K shell) electrons are not involved in the bonding (i.e., as nonbonding electrons). The valence electrons of the two Li atoms are used to populate a new σ_{2s} MO as shown in Fig. 2.42. Representing each of the filled $1s$ orbitals by K (for a K shell) the electronic configuration of Li₂ can be represented as;



The configuration of Li₂ is much like that of H₂, and the bond order which can be determined from the valence electrons only, is equal to $\frac{1}{2} (2-0)$, or 1.

With a bond order of 1 the Li₂ molecule is predicted to exist. Neither liquid nor solid Li consists of Li₂ molecules, but diatomic molecules are indeed found in gaseous lithium. The bond energy in Li₂ is 105 kJ mol^{-1} . This is lower than in H₂ (433 kJ mol^{-1}) because of the shielding of the nucleus by the complete K shell in each atom.

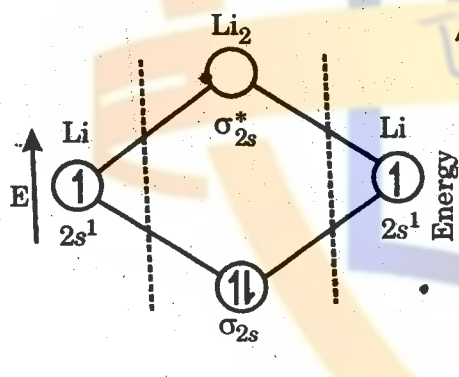


Fig. 2.42. Formation of MOs for Li₂

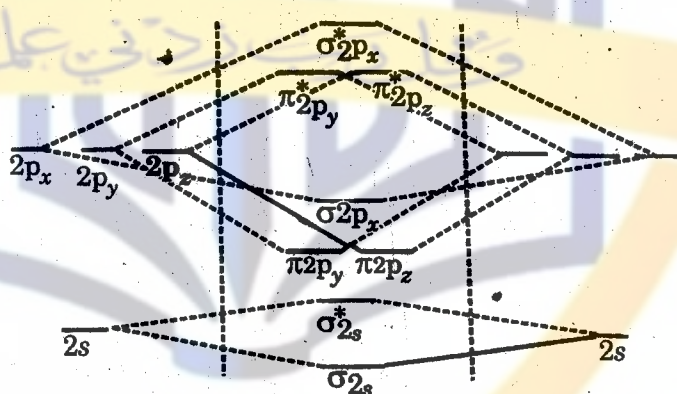


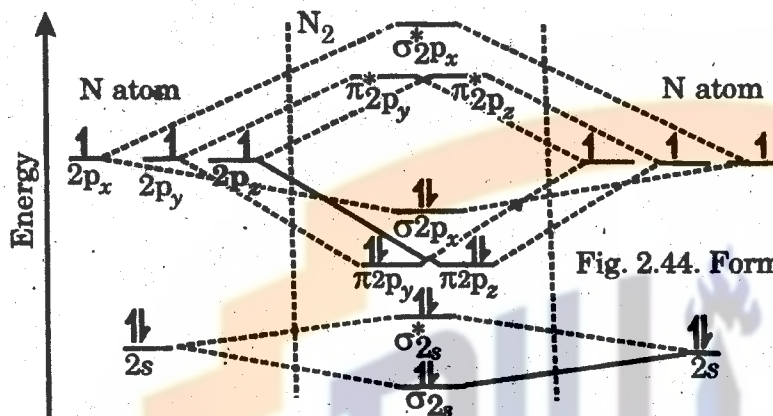
Fig. 2.43. Molecular energies: B₂, C₂ and N₂

N₂ Molecule: The electronic configuration of each nitrogen atom ($Z = 7$) is $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$. The inner shell (K - shell) electrons do not take part in bonding. Therefore, two $2s$ and three $2p$ electrons from each nitrogen atom are to be considered in the bonding interactions. The formation of N₂ proceeds according to the following equation;



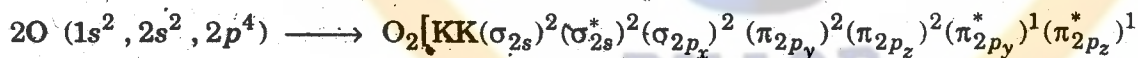
The total number of valence electrons in bonding orbitals is eight and the no. of electrons in antibonding orbital is two. Thus, N₂ has a net excess of six bonding electrons, which corresponds to a bond order of 3 (triple bond).

$$\begin{aligned} \text{Bond order} &= \frac{\text{No. of } e^- \text{ in bonding MOs} - \text{No. of } e^- \text{ in antibonding MOs}}{2} \\ &= \frac{8-2}{2} = \frac{6}{2} = 3 \end{aligned}$$

Fig. 2.44. Formation of N_2 molecule

N_2 is very stable since it has the largest excess of bonding over antibonding electrons. Magnetic measurements confirm that all electrons are paired in N_2 : it is not paramagnetic. The six electrons in the π_{2p_y} , π_{2p_z} and σ_{2p_x} orbitals correspond to the six electrons shown in the Lewis structure : $N \equiv N$:

O_2 Molecule. The electronic configuration of each O atom ($Z = 8$) is $1s^2, 2s^2, 2p_x^2, 2p_y^1, 2p_z^1$. The K-shell electrons do not take part in bonding. Therefore, two 2s and four electrons from each oxygen atom are to be considered in the bonding interactions. The formation of O_2 proceeds according to the following equation;



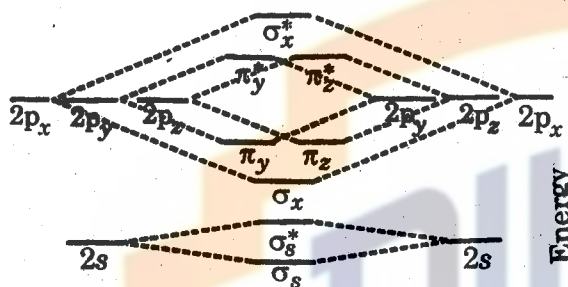
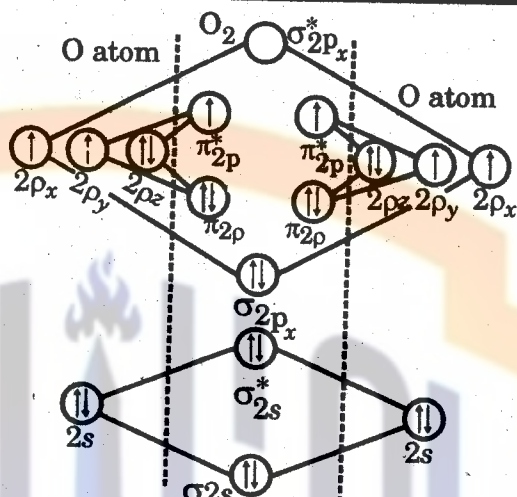
The O_2 molecule contains two electrons more than N_2 molecule and these electrons must go into the degenerate antibonding π_{2p}^* (i.e. $\pi_{2p_y}^*$ and $\pi_{2p_z}^*$) molecular orbitals according to **Hund's rule**.

The total number of valence electrons in bonding orbitals is eight, and the number of electrons in antibonding orbitals is four. Thus O_2 has a net excess of four bonding electrons, which corresponds to a bond order of 2.

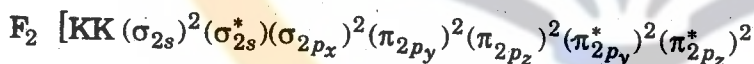
O_2 molecule has two unpaired electrons, even though the total number of electrons is even. Thus the paramagnetism of O_2 is explained.

The lower bond order of O_2 than N_2 is consistent with the fact that O_2 has a smaller bond energy and a longer bond length than does N_2 .

The MO theory readily accounts for the observed magnetic and bond characteristics.

Fig. 2.45. MO energies: O_2 , F_2 and Ne_2 Fig. 2.46 Formation of O_2 molecule.

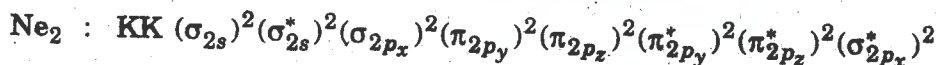
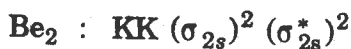
F_2 molecule. The addition of two more electrons to the O_2 configuration gives the orbital population of F_2 . Note that these two electrons must go into the two degenerate antibonding (π_{2p}^* (i.e., ($\pi_{2p_y}^*$ and $\pi_{2p_z}^*$)) orbitals. Since the antibonding π_{2p}^* orbitals are both filled, the bond order in F_2 is only 1. The effective F-F bonding is provided by a σ bond due to the two electrons present in the σ_{2p} MO. All the other electrons in bonding MOs are cancelled by those in corresponding antibonding MOs. The electronic configuration in F_2 molecule is;



Cl_2 and Br_2 have analogous structures. Chlorine has its K- and L- shells and Bromine has its, K, L and M shells full or non-bonding.

Example 2.2: Would you expect Be_2 and Ne_2 exist as stable molecules?

Solution: A stable diatomic molecule must have an excess of bonding over antibonding electrons. The electronic configuration of Be_2 and Ne_2 are;



Both Be_2 and Ne_2 have equal numbers of bonding and antibonding electrons and a bond order of zero. They do not exist as stable molecules. Indeed Be_2 and Ne_2 have never been observed

2.16 HETERONUCLEAR DIATOMIC MOLECULES

Diatomic molecules having different atoms are called heteronuclear. The difference in electronegativities in these molecules causes the MO energy spacings to be different from those in homonuclear diatomics.

The general conditions for the most effective combination of atomic orbitals in a molecule are as follows:-

The AOs involved should: (i) have similar energies; (ii) overlap as much as possible; and have the same symmetry with respect to the x (i.e. internuclear) axis.

In a heteronuclear molecule AB the choice of AOs for combination is guided by information from atomic spectroscopy. The molecular wave function ψ obtained by the linear combination of $1s$ -orbitals, on two dissimilar atoms can be written as;

$$\psi = C_A 1s_A + C_B 1s_B$$

Where the weighting coefficients are unequal. The electron distribution is given by the square of this wave function;

$$\psi^2 = C_A^2 (1s_A)^2 + C_B^2 (1s_B)^2 + 2C_A C_B (1s_A)(1s_B)$$

If C_B^2 is greater than C_A^2 there is a greater probability of finding the electron in the orbital of atom B than in that of atom A.

Let us now apply the MO theory to explain the bonding in heteronuclear diatomic molecules such as HF. The electronic configuration of H and F atoms are;



Combinations of the hydrogen $1s$ orbital with the inner-shell orbital $1s$ or even the $2s$ orbital of F can be ruled out, because the energies of these orbitals of F are much too low. The $2p$ orbitals of F have suitable energy and they are involved in bond formation. Let us consider which of the $2p$ orbitals is able to overlap more effectively with H ($1s$) AO so as to form available MO.

The MO are classified into σ and π type. The hydrogen $1s$ orbital is of σ type in the context of molecular symmetry and so can combine with the $2p$ orbital of the F atom, if we take the molecular (internuclear) axis as the x -axis. The $2p_y$ and $2p_z$ orbitals are of π type, and so do not take part in the bonding because there are no hydrogen orbitals of π type except at very high energy. The $2p_x$ atomic orbital of F combines with $1s$ atomic orbital of H to form an effective σ -bond overlap as shown in Fig. 2.47.

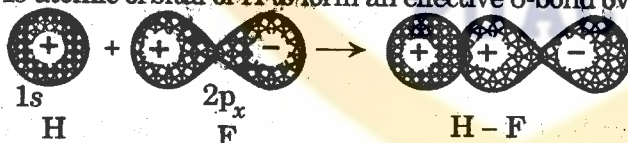


Fig. 2.47 Combination of $1s$ -orbital of H and $2p_x$ orbital of F to form HF.

On the other hand, if $2p_y$ or $2p_z$ orbitals of F overlap with $1s$ atomic orbital of H broadside on, the overlap from the positive lobe will be counterbalanced by the negative lobe as shown in Fig. 2.48. It may be now concluded that only $2p_x$ atomic orbital is responsible for the bond formation between H and F.

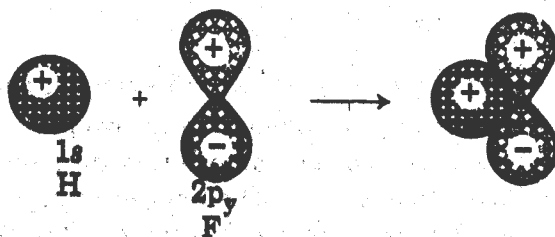


Fig. 2.48. Vanishing of the overlap by H ($1s$) orbital with F($2p_y$) orbital

Thus only the $2p_x$ orbital of fluorine atom has proper symmetry and energy to interact with the $1s$ orbital of the H-atom, giving one sigma bonding molecular orbital and one sigma antibonding molecular orbital. The $2p_y$ and $2p_z$ orbitals of fluorine yield non-bonding molecular (NB) orbitals at the same energies because of their symmetry they yield zero overlap with $1s$ orbital. These are actually unchanged atomic orbital of fluorine atom.

The MO is σ_x which is of σ -symmetry along the internuclear x -axis.

The molecular orbital diagram is shown in Fig. 2.49. The $2p_x$ orbitals of F are shown at a relatively lower energy state because of greater electronegativity of F. The energy difference 'e' between AOs corresponds to the extent of ionic character in HF which depends upon the difference in electronegativities between two atoms. The covalent character of bond is determined by the energy difference 'C' and would indicate the extent of overlap. The bond order in HF is 1. Therefore, the MO theory is able to predict the nature of the chemical bond to be expected from a molecule.

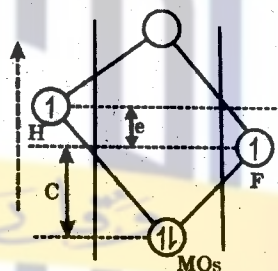


Fig. 2.49 Formation of HF

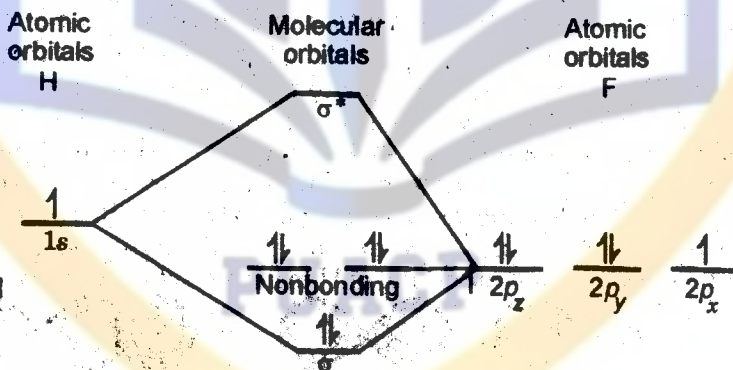


Fig. 2.50 Molecular orbital diagram for the $1s$ atomic orbital of hydrogen and the $2p$ atomic orbitals of fluorine in the HF molecule.

Carbon monoxide Molecule, CO

The electronic configurations of participating C and O atoms are:

$$C = 1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^1; \quad O = 1s^2, 2s^2, 2p_x^2, 2p_y^1, 2p_z^2$$

Evidently CO molecule has 10 valence electrons: four from C atom ($2s^2 2p^2$) and six from O atom ($2s^2 2p^4$) and therefore, its molecular orbital configuration is

$$CO = KK (\sigma_{2s})^2, (\sigma_{2s}^*)^2, (\pi_{2p_y})^2 = (\pi_{2p_z})^2, (\sigma_{2p_x})^2$$

Thus its bond order is equal to $\frac{1}{2} (8-2) = 3$ showing that C and O atoms in CO molecule are linked together by one sigma (σ) and two pi (π) bonds.

A simplified diagram of the molecular orbitals derived from the $2s$ and $2p$ atomic orbitals is shown below in Fig 2.51. The oxygen atomic orbitals are lower in energy than those of carbon as a result of the greater Z_{eff} , but they are close enough in energy that we can construct a molecular orbital diagram similar to that of

homonuclear diatomic molecule. A major difference between homonuclear and heteronuclear diatomic molecule is that the molecular orbitals derived primarily from the 2s atomic orbitals of one element overlap significantly in energy with those derived from the 2p atomic orbitals of the other element. Thus we must consider molecular orbitals derived from both these atomic orbitals in our diagram. Furthermore, because of the asymmetry of orbital energies, the bonding molecular orbitals are derived mostly from the oxygen atomic orbitals, whereas the antibonding molecular orbitals are derived mostly from the carbon atomic orbitals. Finally, there are two molecular orbitals whose energies are between those of contributing atomic orbitals. These orbitals, σ^{NB} , are defined as nonbonding, that is, they do not contribute significantly to the bonding. To determine the bond order of CO, the number of antibonding pairs (0) is subtracted from the number of bonding pairs (3), a calculation leading to the prediction of a triple bond. This very high bond energy of 1072 kJ mol^{-1} supports this prediction.

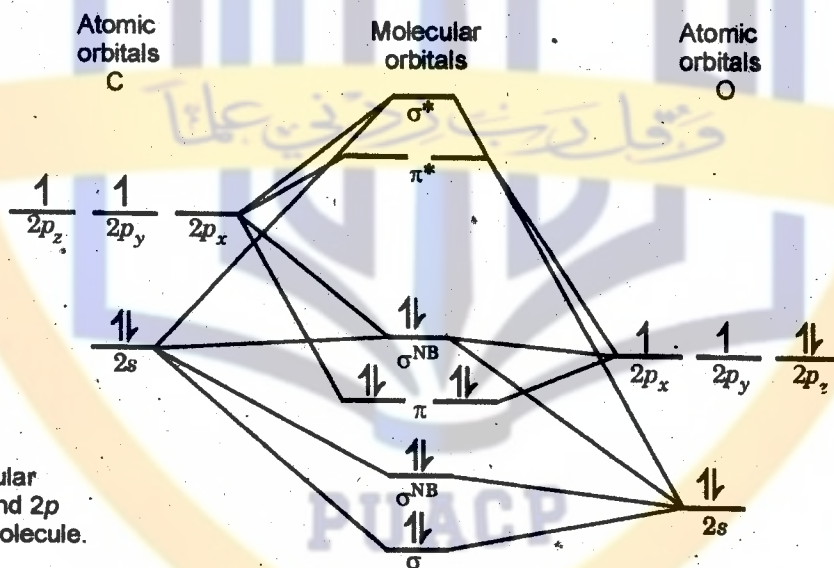


Fig. 2.51 Simplified molecular orbital diagram for the 2s and 2p atomic orbitals of the CO molecule.

Nitric oxide molecule (NO). NO molecule has 11 electron in its valence shell: five from N-atom ($2s^2p^3$) and six from O atom ($2s^2p^4$). Thus it has the following molecular orbital configuration:

$$(\sigma_{2s}^b)^2, (\sigma_{2s}^{*b})^2, (\pi_y^b)^2 = (\pi_z^b)^2, (\sigma_x^b)^2, (\pi_y^*)^0 = (\pi_z^*)^1$$

The bond order equal to $(8-3)/2 = 5/2$ shows that nitrogen and oxygen in the molecule are linked together by one σ and two π -bonds. σ -bond is due to the electron pair in σ_x^b molecular orbital and the two π -bonds are due to π_y^b and π_z^b molecular orbitals. One unpaired electron is present in π_z^* molecular orbital, which explains the paramagnetic character of NO. Thus, NO molecule has two 2-electron bonds and one 3-electron bond and can therefore be represented as $\text{:N} \equiv \text{O} \cdot$.

This structure well accounts for the lower bond strength and larger bond length as compared with N_2 molecule. See Fig in oxides of nitrogen; NO.

2.17 Molecular Orbital Theory for Polyatomic molecules

Molecular orbital theory can also be used to develop bonding schemes for polyatomic molecules. However energy diagrams and the orbital shapes become more and more complex. In general, a molecular orbital in a polyatomic system extends over all the nuclei in a molecule. Indeed MOT forms the basis for most of the quantitative investigations of the properties of large molecules. Let us consider the simplest **linear triatomic molecule**; BeH_2 . Let us choose the x axis as the molecular axis. Note that

only σ molecular orbitals can be formed because H atoms have only their $1s$ orbitals to use in bonding. These orbitals are themselves of σ character with respect to any axis that passes through the nucleus, and therefore they can contribute only to σ molecular orbitals. Then, on the Be atom, only the $2s$ and $2p_x$ orbitals can participate in bonding. The p_y and p_z orbitals, which have π character and zero overlap with any σ orbital, will not play a role in bonding in the BeH_2 molecule.

The $2s$ of Be can combine with the two $1s$ orbitals of the H atoms to form bonding and antibonding MO's, as is shown in Fig. 2.52.

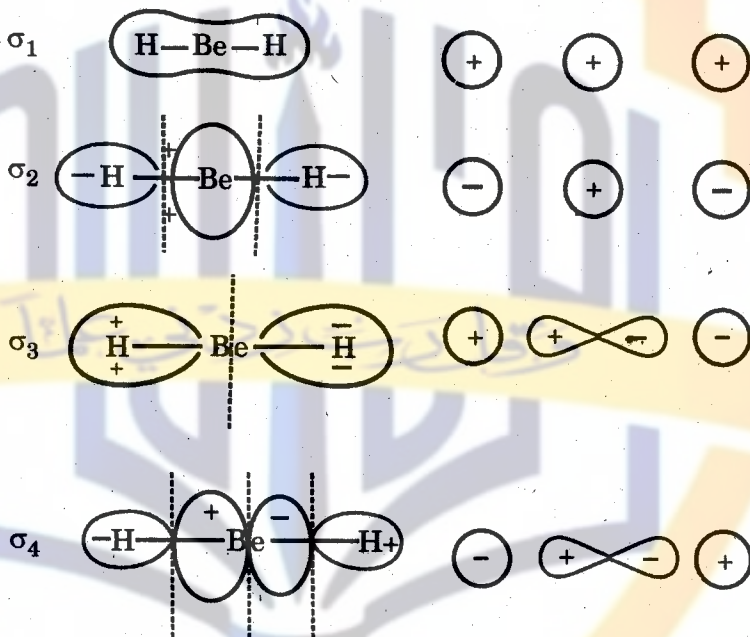


Fig. 2.52 The four σ molecular orbitals for the linear BeH_2 . The dashed vertical lines are nodal planes perpendicular to the molecular axis. Those p atomic orbitals of Be that are perpendicular to the molecular axis have nonbonding interactions with the hydrogen $1s$ orbitals.

The $2p_x$ orbital of Be also combines with the hydrogen $1s$ orbitals, as is shown in Fig. 2.52 to form bonding and antibonding molecular orbitals. In these, $1s$ orbitals are out of phase with each other.

The important points to remember about these four σ molecular orbitals are as follows:

1. In each bonding MO, electron density is large and continuous between adjacent atoms, while in the antibonding MO's there is a node between each adjacent pair of nuclei.
2. Electrons in MOs are *delocalized* over the whole extent of the MO.

The MO treatment of the bonding in BeH_2 can be expressed in terms of an energy-level diagram as shown in Fig. 2.53. The main points here are that the H 1s orbitals lie at much lower energy (-400 kJ/mol) than the Be 2s orbital and that the p_y and p_z orbitals of Be remain completely unchanged into centre column, because they do not overlap with any other orbitals. The four valence electrons, $2s^2$ from Be and 1s from each H, occupy σ_1 and σ_2 . The total bond order of the Be-H bonds is two. Since each Be-H pair participates equally in the molecules, this is equivalent to saying that there are two equivalent Be-H single bonds.

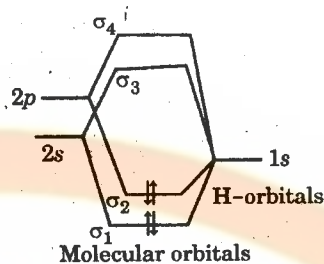


Fig. 2.53 MO energy level diagram for BeH_2 . The MO designations σ_1 – σ_4 correspond to those of above Fig. 2.52.

Structure of Water, H_2O

An analysis of the molecular orbitals for the H_2O molecule provides a good introduction to the way in which the symmetry of a molecule determines the forms of the molecular orbitals in a polyatomic system. There are three symmetry operations, which transform the nuclear framework of the water molecule into itself. For each symmetry operation, there is a corresponding symmetry element. The symmetry elements for the H_2O molecule are a two-fold axis of rotation C_2 and two planes of symmetry σ_1 and σ_2 . A rotation of 180° about the C_2 axis

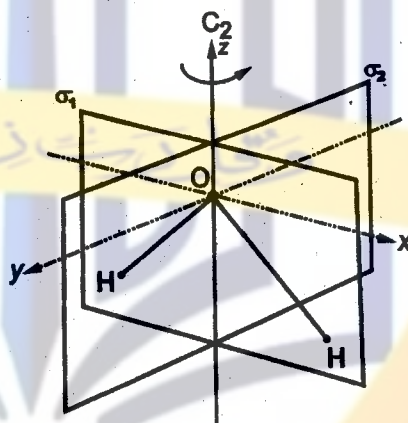


Fig. Symmetry elements for H_2O

leaves the oxygen nucleus unchanged and interchanges the two hydrogen nuclei. A reflection through the plane labelled σ_1 leaves all the nuclear positions unchanged while a reflection through σ_2 interchanges the two protons. The symmetry operations associated with the three symmetry elements either leave the nuclear positions unchanged or interchange symmetrically equivalent (and hence indistinguishable) nuclei. Every molecular orbital for the H_2O molecule must, under the same symmetry operations, be left unchanged or undergo a change in sign.

We may use the symmetry transformation properties of the atomic orbitals on oxygen and hydrogen together with their relative orbital energy values to determine the primary atomic components of each molecular orbital in a simple LCAO approximation to the exact molecular orbitals. Only atomic orbitals which transform in the same way under the symmetry operations may be combined to form a molecular orbital of a given symmetry. The symmetry transformation properties of the atomic orbitals on oxygen and hydrogen are given in table. A value of +1 or -1 opposite a given orbital in the table indicates that the orbital is unchanged or changed in sign respectively by a particular symmetry operation.

Table: Symmetry Properties

Atomic orbitals on oxygen	Symmetry behavior under σ_2 σ_1 σ_2	Symmetry classification	Atomic orbitals on H	Symmetry behaviour under σ_2 σ_1 σ_2	Symmetry classification
1s	+1 +1 +1	a_1	$(1s_1 + 1s_2)$	+1 +1 +1	a_1
2s	+1 +1 +1	a_1	$(1s_1 - 1s_2)$	-1 +1 -1	b_1
$2p_z$	+1 +1 +1	a_1			
$2p_x$	-1 +1 -1	b_2			
$2p_y$	-1 -1 +1	b_1			

The 1s, 2s and $2p_z$ orbitals of oxygen are symmetric (i.e., unchanged) with respect to all three symmetry operations. They are given the symmetry classification a_1 . The $2p_x$ orbital, since it possesses a node in the σ_2 plane changes sign when reflected through the σ_2 plane or when rotated by 180° about the C_2 axis. It is classified as b_2 orbital. The $2p_y$ orbital is antisymmetric with respect to the rotation operator and to a reflection through the σ_1 plane. It is labelled b_1 .

The hydrogen 1s orbitals when considered separately are neither unchanged nor changed in sign by the rotation operator or by a reflection through the σ_2 plane. Instead both these operations interchange these orbitals. The hydrogen orbitals are said to be symmetrically equivalent and when considered individually they do not reflect the symmetry properties of the molecule. However, the two linear combinations $(1s_1 + 1s_2)$ and $(1s_1 - 1s_2)$ do behave in the required manner. The $(1s_1 + 1s_2)$ is symmetric under all three operations and is of a_1 symmetry while the $(1s_1 - 1s_2)$ is antisymmetric with respect to rotation operator and to reflection through the plane σ_2 and is of b_2 symmetry.

The molecular orbitals in H_2O molecule are classified as a_1 , b_1 or b_2 orbitals, as determined by their symmetry properties. In addition to the symmetry properties of the atomic orbitals we must consider their relative energies to determine which orbitals will overlap significantly and form delocalized molecular orbitals.

The 1s atomic orbital on oxygen possesses a much lower energy than any of the other orbitals of a_1 symmetry and should not interact significantly with them. The molecular orbital of lowest energy in H_2O should therefore correspond to an inner shell 1s atomic like orbital centered on the oxygen. This is the first orbital of a_1 symmetry and it is labeled $1a_1$. Notice that the orbital energy of the $1a_1$ molecular orbital is very similar to that for the 1s atomic orbital on oxygen.

The atomic orbital of next lowest energy in this system is the $2s$ orbital of a_1 symmetry on oxygen. This orbital will overlap with the $(1s_1 + 1s_2)$ combination of orbitals on the H atoms to form the $2a_1$ molecular orbital and is a binding orbital.

Aside from the presence of the second proton, the general form and nodal structure of the $2a_1$ density distribution in the H_2O molecule is remarkably similar to the 2σ distributions in HF molecule.

Although the three $2p$ atomic orbitals are degenerate in the O atom, the presence of the two protons results in each $2p$ orbital experiencing a different potential field in the H_2O molecule. The nonequivalence of the $2p$ orbitals in the H_2O molecule is evidenced by all three possessing different symmetry properties. The three $2p$ orbitals will interact to different extents with the protons and their energies will differ.

The $2p_x$ orbital interacts with the protons and forms an orbital of b_2 symmetry by overlapping with the $(1s_1 - 1s_2)$ combinations of $1s$ orbitals on the hydrogens and the result is the $1b_2$ molecular orbital. The $1b_2$ orbital concentrates charge density along each O-H bond axis and draws the nuclei together. The charge density of $1b_2$ orbital binds all the three nuclei.

The $2p_z$ orbital may also overlap with the H $1s$ orbitals, the $(1s_1 + 1s_2)$ a_1 combination, and the result is the $3a_1$ molecular orbital. This orbital is concentrated along the z -axis and charge density is accumulated in both the bonded and non-bonded sides of the O nucleus. It exerts a binding force on the protons and an anti-bonding force on the oxygen nucleus.

The $2p_y$ orbital is not of the correct symmetry to overlap with the hydrogen $1s$ orbitals. The $1b_1$ molecular orbital will be simply a $2p_y$ atomic orbital on the O, perpendicular to the plane of the molecule. The electronic configuration of the H_2O

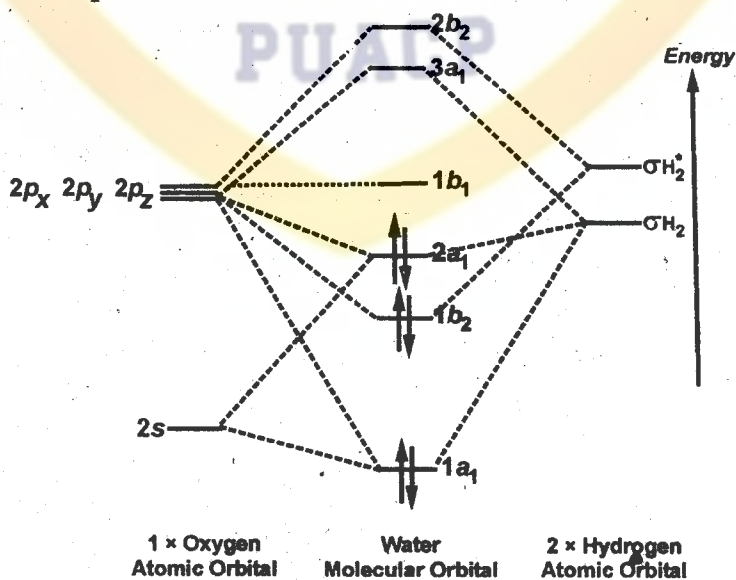


Fig. H_2O Molecule

molecular as determined by MOT is: $1a_1^2, 2a_1^2, 1b_2^2, 3a_1^2, 1b_1^2$.

Trigonal planar Molecules: AB_3

An important and more general application of MO theory in polyatomic molecules deals with π bonding in planar systems. One important group of the symmetrical compounds of the general formula AB_3 , e.g., BF_3 , CO_3^{2-} and NO_3^- . If these trigonal planar systems are oriented so that the central atom is at the origin of the coordinate system and the molecular or ionic plane coincides with the xy plane, then the π bond system will be formed entirely by the p_x atomic orbitals of the four atoms. The π -bond system must then have a node in the xy plane. It must also be equally dispersed over the three A-B bonds, in agreement with the resonance result. Consistent with this, the MO approach involves overlap that encompasses the whole structure.

There are three linear combinations of p_x atomic orbitals from the peripheral B atoms of AB_3 . We shall not discuss the methods that are used to deduce these particular linear combinations, except to point out that these three particular linear combinations have been chosen to provide the best total, positive overlap with the atomic p_x orbital of the central atom A. These three linear combinations are termed **group orbitals** as shown in Fig. 2.54. The linear combinations [group orbitals (GO)] represented in the Fig. 2.54 arise from three distinct arrangements of the p_x atomic orbital at the vertices of the AB_3 triangle.

Only one group orbital of Fig. 2.54 has non-zero overlap with the p_x atomic orbital of the central atom A as shown in Fig. 2.55. The other two interactions are non-bonding (e.g., π_{2a} and π_{2b}) and the resultant MO's appear in the energy-level diagram of Fig. 2.56 with energy that are unchanged. One of the interactions shown in Fig. 2.55 is bonding and this leads to the π_1

molecular orbital shown in Fig. 2.55. Its antibonding counterpart is shown in Fig. 2.56 as π_1^* .

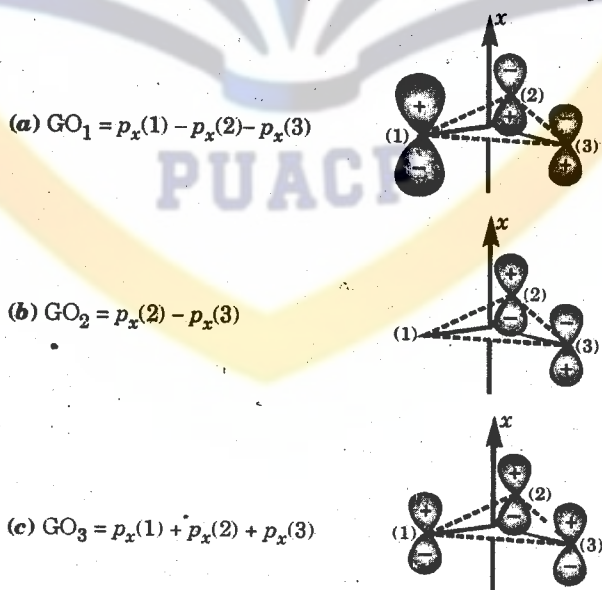


Fig. 2.54. Three combinations of p_x orbitals from the three outer B atoms of a planar AB_3 molecule. Each combination (called a group orbital GO) is multicentered and each is constructed to be used as a group in overlapping with the p_x atomic orbital of the central atom A, as shown in Fig. 2.55.

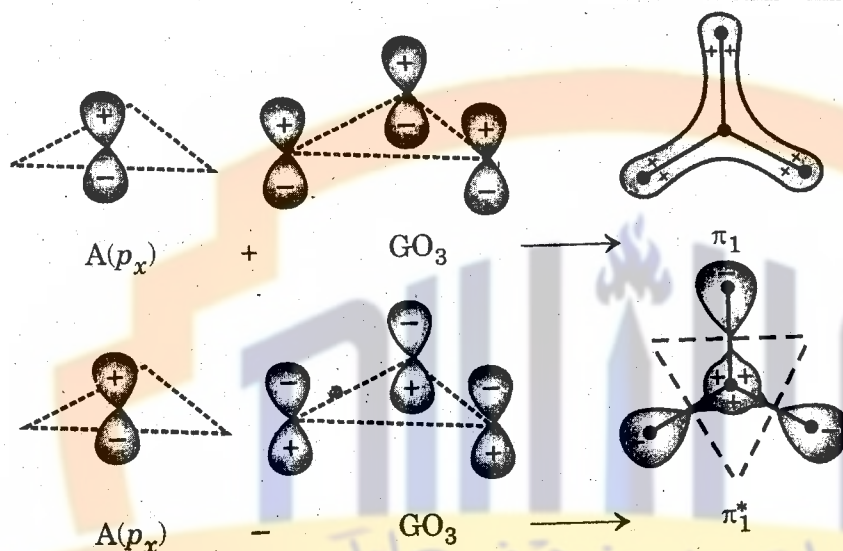


Fig. 2.55. Diagrams showing how bonding (π_1) and antibonding (π_1^*) molecular orbitals are formed in an AB_3 molecule using the p_x orbital of the central atom A and a group orbital (GO_3 of Fig. 2.54) from the outer atoms B. The MO's themselves, at the right, are viewed from above. The MO's change sign in the molecular plane, as do the p_x atomic orbitals from which they are formed. In addition, π_1^* has three nodal planes perpendicular to the molecular plane.

For each of the species BF_3 , CO_3^{2-} ,

or NO_3^- , there are only 6 electrons to occupy the π molecular orbitals of Fig. 2.56 (The other 18 valence electrons occupy the various σ orbitals of the yz plane). The six π electrons are distributed as shown in Fig. 2.56. The 4 electrons in the two degenerate orbitals (π_{2a} and π_{2b}) neither contribute to nor detract the stability of the π bond system because they are nonbonding. Thus the π -bonding stability is provided entirely by the one electron pair in the π_1 molecular orbital. The total π

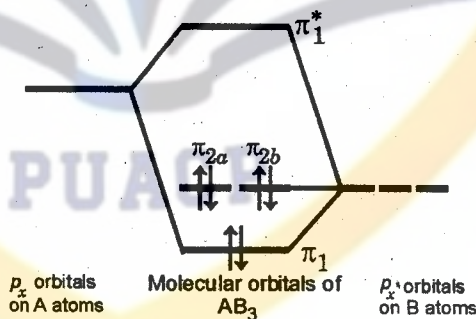


Fig. 2.56. The MO energy-level diagram for the p-bond system in a planar, symmetrical AB_3 molecule. The symbol π_1 and π_1^* correspond to those used in Fig. 2.55. The nonbonding orbitals π_{2a} and π_{2b} are essentially GO_1 and GO_2 of Fig. 2.54.

bond order of one is equally distributed over the three equivalent AB regions so that the net π bond per AB group is one-third. The conclusion here is that one π bond is delocalized over three AB pairs.

Example 2.3:

What is the form of the wave function that describes a molecule in which an electron spends 90 per cent of its time in an orbital ψ_A on A and 10 percent in ψ_B on B in the polar molecule AB?

Solution. The wave function is written in the form:

$$\psi = C_A \psi_A + C_B \psi_B + \text{such that } [C_A]^2 = 0.90 \text{ and } [C_B]^2 = 0.10$$

$$\text{Since } [C_A]^2 = 0.90, \quad C_A = \pm 0.95 \text{ and}$$

$$\text{Since } [C_B]^2 = 0.10, \quad C_B = \pm 0.32$$

For the bonding orbital we take the in-phase overlap so that $\psi = 0.95 \psi_A + 0.32 \psi_B$.

Note: The squares of the coefficients determine the probabilities.

2.18 COMPARISON OF VB AND MO THEORIES

Both the VB and MO theories have been used for the purpose of describing the covalent bond, both employ quantum mechanics. The overlap criterion is predominant in both concepts. The VB theory does not provide the clear picture of the energy changes involved during bond formation. Molecular orbital approach is able to indicate even the minor energy changes during the formation of a molecule from which a lot of information may be collected, the contribution of covalent and ionic character etc. Valence bond theory is useful in predicting the symmetry of the molecules using the idea of hybridization. The most common features between the two theories are:

1. The electron distributions for a given molecule are similar.
2. In both the theories, a normal covalent bond involves the sharing of electron charge density between the nuclei.
3. In both the theories, σ and π bonds can be recognized.
4. In both the approaches, AOs of the atoms must overlap significantly and should have appropriate symmetry about the molecular axis to form a bond.
5. Variation method is used in both the treatments.
6. The variation method is a systematic effort to determine the electron distribution in a particular atom or molecule.

THE ESSENTIAL POINTS OF DIFFERENCE BETWEEN THE TWO THEORIES

1. The basic assumptions of the two theories are quite different. VB theory considers that atoms exist within molecules, and that the structure of a molecule can be interpreted in terms of its constituent atoms and the bonds between them. Whereas, the MO theory, assumes that the atomic orbitals of the original unbounded atoms become replaced by a new set of molecular energy levels, called molecular orbitals and these MOs determine the properties of the resulting molecule.
2. VB theory utilizes the concept of resonance, whereas the MO theory gives an idea of the delocalization.
3. The MO theory gives equal weight to the ionic and covalent terms, whereas, the VB theory shows hundred percent covalent character.

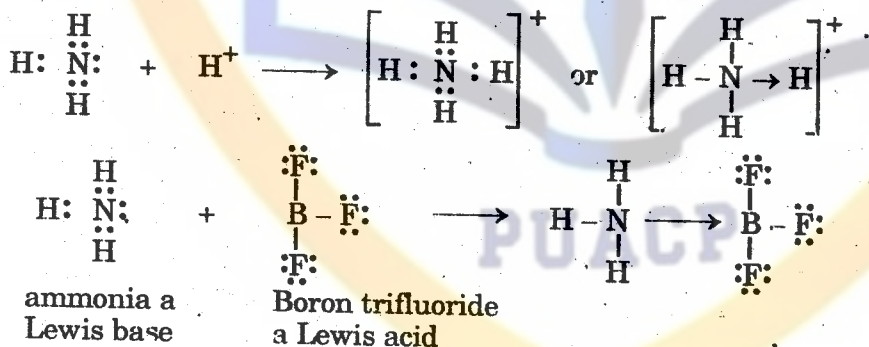
$$\psi_{MO} = \psi_{\text{ionic}} + \psi_{\text{cov}}$$

$$\psi_{VB} = \psi_{\text{cov}}$$

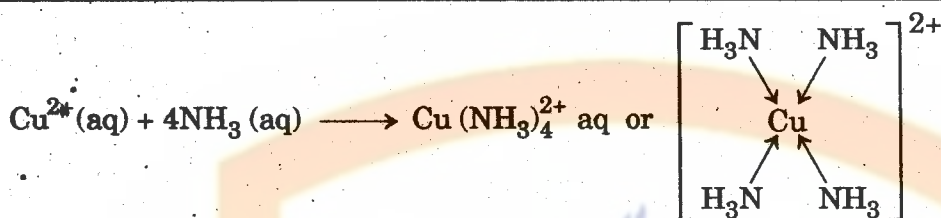
4. The excited states in molecules can be easily described by MO theory, whereas excited states cannot be easily described by the VB theory.
5. The MO theory accounts for the observed paramagnetic character of O_2 but the VB theory is unable to explain the paramagnetic character of O_2 .
6. MO theory accounts quite nicely for one-electron bonds, which are known to exist, e.g; He_2^+ whereas the VB theory requires electron pair bonds.

2.19. Coordinate covalent bond

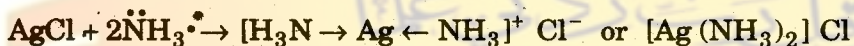
A covalent bond in which the shared electron pair is donated by only one of the two bonded atoms is called coordinate covalent bond. The atom which donates the electron pair is called the "donor" and the atom which accepts them is called the "acceptor". This bond is generally represented by an arrow ($\longrightarrow$) which runs from the donor atom to acceptor. This type of bond is formed between atoms which are either electron-pair donor or electron-pair acceptors. The electron-pair donor atoms are usually called Lewis bases and the electron-pair acceptors are called Lewis acids. Lewis acid-base reactions would result in the formation of coordinate covalent bonds. For example, formation of NH_4^+ ion by the reaction of NH_3 (Lewis base) and proton, H^+ (Lewis acid)



A Lewis base in a coordination compound is called Ligand. Most ligands are anions or substances that contain nonbonding valence electron pairs. Several hundred different ligands are known. Those most commonly encountered in general chemistry are NH_3 and H_2O molecules and halide ions (F^- , Cl^- , Br^- , I^-) and OH^- ions. The most common acceptors are electron deficient molecules such as BF_3 , BCl_3 , AlF_3 , $GaCl_3$ or transition metals having partially or incompletely filled d orbitals. Here transition metals would act as Lewis acids because they are short of electrons due to presence of unfilled d or f orbitals. A typical pair acceptor is H^+ . For example, A water solution containing the Cu^+ ions has a pale blue colour. When aqueous ammonia, NH_3 , is added, the colour changes to a deep, almost opaque, blue due to the formation of $Cu(NH_3)_4^{2+}$ ion which is commonly referred to as a complex ion.



An ion such as $\text{Cu}(\text{NH}_3)_4^{2+}$ cannot exist by itself in the solid state. The +2 charge of this ion must be balanced by anions with a total charge of -2. A typical compound containing the $\text{Cu}(\text{NH}_3)_4^{2+}$ is $[\text{Cu}(\text{NH}_3)_4]\text{Cl}_2$. Compounds such as this which contain a complex ion are referred to as coordination compounds. The formula of the complex ion is set off by brackets, [], to make the structure of the compound clear. Similarly, silver chloride dissolves in ammonia to form the coordination compound $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$ or its ion $[\text{Ag}(\text{NH}_3)_2]^+$



It should be noted that once the coordinate covalent bond is formed, it is indistinguishable from an ordinary covalent bond.

2.20 THE METALLIC BOND

The existence of ionic bonds in metals is obviously impossible since there are only one type of atoms. The high coordination numbers (8 or 12) and presence of quite insufficient electrons for bonding in the metal atoms do not favour the possibility of the existence of covalent bonds between metallic atoms. On the other hand, the cohesive force is too large to be attributed to the weaker van der Waals forces between the metallic atoms.

Bonding in metals due to the electrical attractions among positively charged metal ions and mobile, delocalized electrons belonging to the crystal as a whole is called metallic bonding. In other words, "the force that holds metal atoms together in the crystal is known as the metallic bond". Any theory that attempts to describe the metallic bond must account for the strength of this force and these distinctive properties that all metals share, more or less:

1. Ability to conduct electricity.
2. Ability to conduct heat.
3. Ease of deformation i.e., ability to be flattened into sheets (malleability) and to be drawn into wires (ductility).
4. Lustrous appearance i.e., metal surfaces, when clean, are shiny. The metallic luster results from the reflection of all wavelengths of visible light from the surface of most metals.

Many theories have been proposed to account for these properties for bonding in metals, some of which are described below:

A. Electron Sea or Electron Gas Theory

According to electron gas theory, each atom in a metal crystal loses all its valence electrons. The electrons thus obtained form a *sea* or *gas* and the resulting positively charged metal ions are held together by this *electron sea* or *gas*. The positively charged metal ions do not float randomly in the sea of electrons but have definite position at measurable distances from each other in the crystal lattice. Since the valence electrons are not bound to any individual ions or pairs of ions but are delocalized over the entire crystal, they are therefore called free electrons. The metal is thus pictured as a network or lattice of positive ions of the metal immersed in a 'sea of electrons' or 'gas of elections'. *The force that binds a metal ion to the mobile electrons within its sphere of influence is known as metallic bond.*

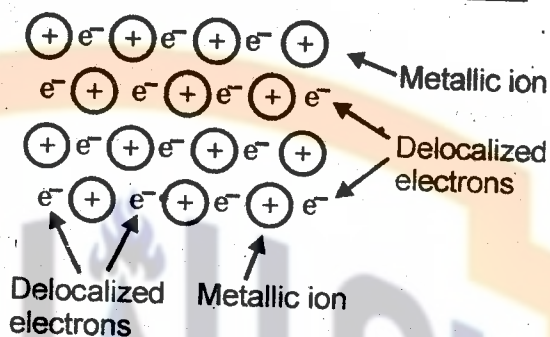


Fig. 2.57 The electron sea model of the metallic bond.

Electron Sea model and metallic properties. The free electrons account for the characteristic metallic properties.

- (i) **Electrical conductivity.** When the ends of a bar of metal are connected to a source of electric current, electrons from the external source enter the bar at one end. Free electrons pass through the metal and leave the other end at the same rate. Passage of electrons from one end to the other constitute electrical conduction,
- (ii) **Thermal conductivity.** When one part of a metal is heated, the electrons in that part acquire a large amount of energy. These electrons, being free, move rapidly through the crystal and conduct heat to the other parts of the metal.
- (iii) **Deformation of metals.** If one layer of metal ions is forced across another, perhaps by hammering, the internal structure remains essentially unchanged as the sea of electrons rapidly adjusts to the new situation and accounts for the ease of deformation of metals.
- (iv) The **lustrous appearance** of a metal is also caused by the free electrons. The unbound electrons at the surface of a metal absorb and radiate light which strikes the surface and reflects light completely, at all angles giving metals their peculiar lustre.
- (v) The **photoelectric effect** and the **thermionic effect** (metal wire emit electrons when heated) can be explained by saying that the ejected electrons jump off the surface of the electron sea because of input of energy. The discovery of the thermionic effect led to the invention of electronic tubes and the birth of the electronic industry.

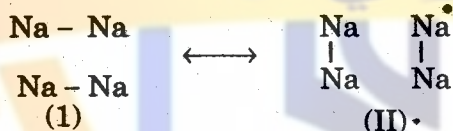
According to the electron-sea model, the strength of the metallic bond is directly related to the charge of positive ions in the lattice. The greater the positive

metal ion, the stronger the forces binding a metal ion to the mobile electrons in the crystal. Therefore, we would expect the "lattice energy" of Mg (+2 ion) to be greater than that of Na (+1 ion). This agrees with the fact that Mg has a higher melting point than Na (650°C vs. 98°C).

It is unreasonable to assume that an electron charge cloud is bound to a single nucleus and is completely delocalized around all the nuclei. The electron sea theory is not very specific about the region occupied by free electrons in a metal.

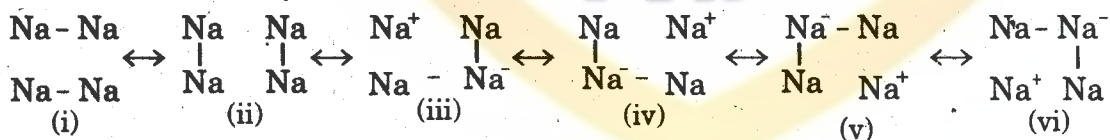
B. Valence Bond Theory (Pauling Theory)

According to Pauling the metallic bonding is essentially covalent in origin and metallic structure involves resonance of electron-pair bonds (i.e., covalent bonds) between each atom and its nearest neighbours. For example, an alkali metal atom has only one valence electron and it can form one covalent bond with only one of its neighbours. Thus in two dimensions, various resonance forms may be depicted as follows.



Here it should be noted that although only four atoms have been shown in these forms, the actual bonding includes all of the atoms of the crystal and is three dimensional.

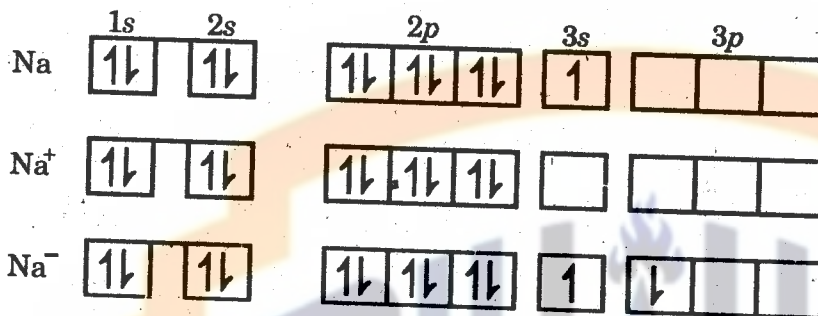
The metallic structure is a resonance hybrid of these forms (I and II). If only these two structure were assumed the covalent bond would relatively be weaker. So in addition to structures I and II the other contributing structures which involve transfer of one electron from one atom to another are also assumed.



Resonance among these six structures would lead to greater stability. The contributing structures (iii), (iv), (v) and (vi) contain a Na-atom (having a negative charge) bonded to other two Na-atoms by covalent bonds. These covalent bonds are resonating covalent bonds. The formation of two covalent bonds shows that sodium has used two orbitals for bonding purpose. The resonance structures can be best considered if unoccupied orbitals of suitable energy are available for hybridization.

The electronic configuration of Na-atom ($Z = 11$) is $1s^2 2s^2 2p^6 3s^1$. Na-atom has four outer orbitals (one 3s and three vacant 3p orbitals). Since the energy of these three 3p-orbitals is not very different from that of the valence 3s orbital, an electron from 3s-orbital of one Na-atom can easily be transferred to one of three 3p vacant orbitals of the other Na-atom and convert it into Na^- ion with the configuration $1s^2, 2s^2, 2p^6, 3s^1, 3p_x^1, 3p_y^0, 3p_z^0$. The Na atom which loses the electron from its 3s

orbital is converted into Na^+ ion with the configuration $1s^2, 2s^2, 2p^6$.



Now $3s^1$ and $3p_x^1$ orbitals on Na^- ion form two sp hybrid orbitals which form two covalent bonds with two different Na atoms. The extra two vacant orbitals namely $3p_y^0$ and $3p_z^0$ on Na^- ion are called metallic orbitals¹ and are responsible for the formation of metallic bonds; that is why Na is metal. On the other hand, carbon is nonmetal because it does not have any vacant orbital after sp^3 hybridization.

The strength of the metallic bonds depends upon the number of valency electrons that each atom can contribute. The greater the metallic valence, the stronger the forces holding the atoms of the crystal together. The metallic valence of Na, Mg and Al are 1, 2 and 3, respectively. Therefore, atoms are held more tightly in solid Al than in Mg and more tightly in Mg than in Na. The vacant orbitals on Na^- can accept the conductivity electrons and thus contribute to metallic conduction.

(c) Molecular orbital theory (Band theory) of metals

The molecular orbital theory of metals is commonly known as band theory. In the band theory all electrons present in an atom in completely filled energy levels are considered to be essentially localized and take no significant part in chemical bonding. On the other hand, the valence electrons in unfilled energy levels are considered to be free and can move throughout the whole of the metal, and are not localized to particular atoms. To some extent, therefore, the atoms in the metal do not effect the properties of the valence electrons which give metal their characteristic properties. The atomic orbitals of these free electrons in one atom can overlap those of the electrons in other atoms to give delocalized MOs which produce bonding all the atoms present. The molecular orbitals of these free electrons are called the conduction orbitals of the metals. Two atomic orbitals overlap to form two molecular orbitals: a bonding orbital and an antibonding orbital.

Let us apply this description to the bonding of the sodium atoms. When the two atoms come together, the $3s$ orbitals (each having one electron) of the atoms interact (overlap) to form two molecular orbitals (one bonding orbital and on

antibonding orbital). The 3s electron of each atom is in the lower-energy bonding molecular orbital and the higher-energy antibonding orbital is empty. If N atomic orbitals interact, N molecular orbitals are formed. In a single metallic crystal containing one mole of Na atoms, for example, the interaction of 6.022×10^{23} 3s atomic orbitals produces 6.022×10^{23} molecular orbitals. Atoms interact more strongly with those nearby than with those farther away. The energy separating bonding and antibonding molecular orbitals resulting from two given atomic orbitals decreases as the interaction (overlap) between the atomic orbitals decreases. Thus, when all possible interactions among one mole of Na atoms are considered, there results a series of very closely spaced molecular orbitals (formally σ_{3s} and σ_{3s}^*) that form a nearly continuous band of orbitals belonging to the crystal as a whole. The 6.022×10^{23} orbitals in the band are half-filled, since one mole of Na atoms contributes 6.022×10^{23} valence electrons (Fig. 2.58).

The empty 3p atomic orbitals of the Na atoms also interact to form a wide band of $3 \times 6.022 \times 10^{23}$ orbitals. Since the 3s and 3p atomic orbitals are quite close in energy, the two bands of molecular orbitals actually overlap as shown in Fig. 2.59. The two overlapping bands contain $4 \times 6.022 \times 10^{23}$ orbitals and only 6.022×10^{23} electrons since each orbital can accommodate two electrons, the resulting combination of bands is only one-eighth full.

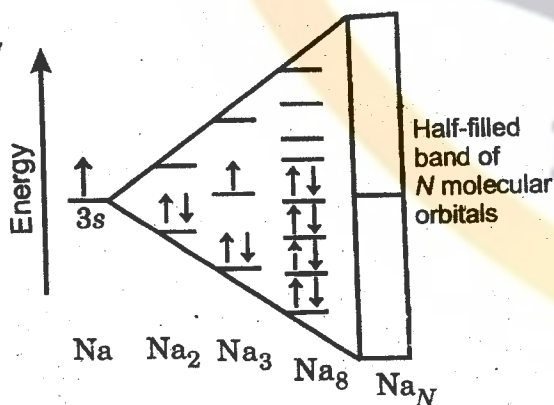


Fig. 2.58. The band of orbitals resulting from interaction of the 3s orbitals in a crystal of sodium.

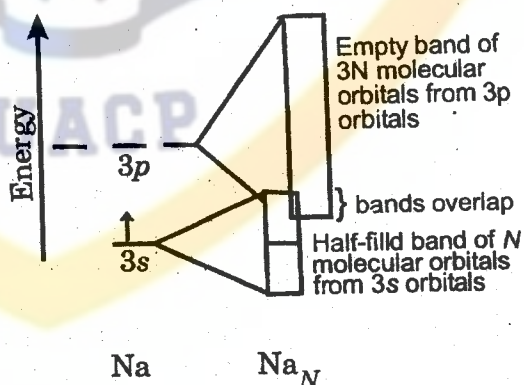


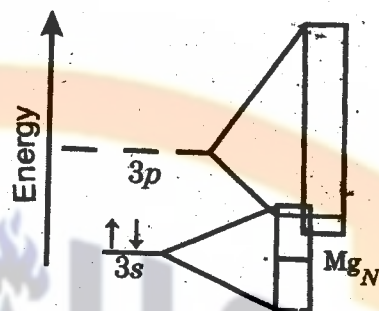
Fig. 2.59. Overlapping of half-filled "3s" band with empty "3p" band of Na_N crystal.

The ability of metallic sodium to conduct electricity is due to the ability of any of the highest energy electrons in the "3s" band to jump a vacant orbital of slightly higher energy in the same band when an electric field is applied. As this occurs, the net flow of electrons through the crystal is in the direction of the applied field.

What about magnesium with its 3s atomic orbital filled with two electrons and therefore, in the absence of this overlap, the "3s" band in a crystal of Mg also would be filled. Why is it not an insulator instead of a conductor. In such cases, the 3p band is low enough to overlap the top of the 3s band, providing a large number of available empty levels (Fig. 2.60) and thus we obtain an unfilled band with free electrons which make Mg a good metallic conductor.

According to band theory, the highest energy electrons of metallic crystals occupy either a partially filled band or a filled band that overlaps an empty band. A band within which electrons must move to allow electrical conduction is called a **conduction band** and can be used to conduct electricity.

Fig. 2.60: Overlapping of filled "3s" band with empty "3p" band of Mg_N crystal. The higher energy electrons are able to move into the "3p" band due to overlap



Crystalline nonmetals, like diamond and phosphorus are insulators because they do not conduct electricity (or conduct heat or have metallic luster). This is because their highest energy electrons occupy filled bands of molecular orbitals that are separated from the lowest empty band (conduction band) by a large energy gap called a forbidden zone, which is too large an energy difference for electrons to jump in order to get the conduction band as shown in Fig. 2.61.

Elements like Ge and Si that are semiconductors have filled bands that are slightly below the empty bands (i.e., conduction bands), but do not overlap with empty bands. They do not conduct electricity at room temperature, but a small increase in temperature is sufficient to excite some of the highest energy electrons into empty conduction band. Once there, these electrons become mobile and impart a degree of conductivity to the metal.

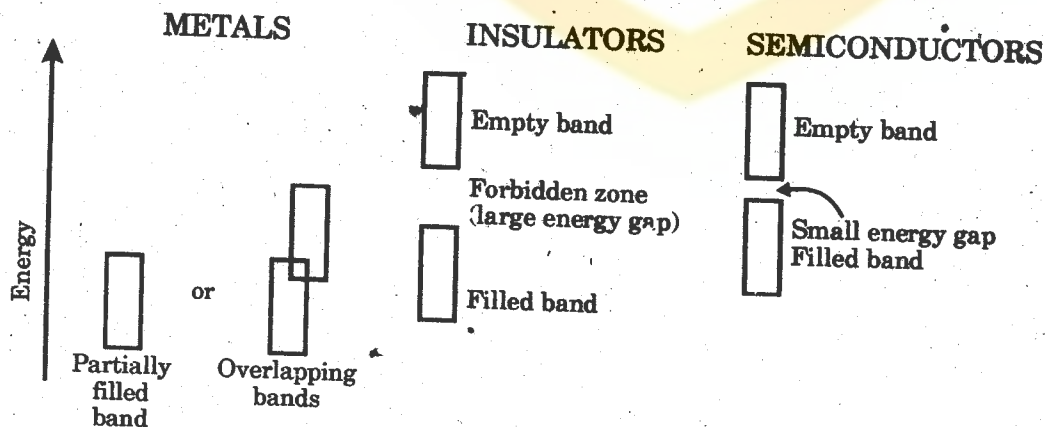


Fig. 2.61. Distinction between metals, insulators and semiconductors.

2.21 van der Waals Forces or Bonds

The concept of van der Waals forces or bonds was introduced to explain the deviation of real gases from ideal behaviour.

In addition to the strong forces that hold atoms together in chemical bonds, there are a number of weaker attractive forces between atoms that are not bonded to each other. Usually, weak interactions are intermolecular—that is, they are attractions between portions of different molecules. These weak intermolecular forces are called van der Waals forces. There are several kind of van der Waals forces, the most important of which are **London forces** and **dipole-dipole interactions** (forces).

London forces, also called **dispersion forces**, are usually very weak. They account for the fact that even nonpolar molecules can form a crystal lattice. These, forces first described theoretically by Fritz London in 1930, arise from momentary asymmetric electron distribution in molecules which produce momentary dipole moments that in turn induce momentary dipole moments in neighbouring molecules; these forces are often called **dipole - induced dipole interaction**.

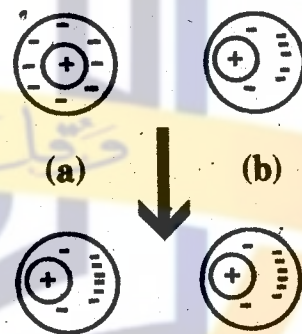


Fig. 2.62 Creation of instantaneous induced temporary dipoles.

London forces depend, in partly, upon the polarizability of the molecule. Several factors influence polarizability. An increase in the number of electrons in the molecule (i.e., increased molecular mass) results in an increase in the polarizability of the molecule. Since high molecular mass indicates stronger the polarizability and therefore, the strength of London forces increase with increased molecular mass. For example, at room temperature, Cl_2 is a gas, Br_2 is a liquid, and I_2 is a solid. The Cl_2 molecule is the smallest of the three and I_2 is the largest. The intermolecular forces between the Cl_2 molecules in the liquid or solid are weaker than the forces between I_2 molecules in the liquid or solid. Another important factor in determining the strength of London forces is molecular shape. Electrons in elongated molecules are more easily displaced than those in small, compact, symmetrical molecules. The elongated normal pentane molecule is more easily polarized than the compact neopentane molecule. Intermolecular forces are stronger in normal pentane than in neopentane.

Increase in melting and boiling points with increase in molecular mass and size may be due to greater energy needed to move a heavy molecule.

Dipole - Dipole interactions

Permanent dipole-dipole interactions occur between polar covalent molecules, because of the attractions of the δ^+ atoms of one molecule to the δ^- atoms of another molecule. Dipole-dipole forces are typically much weaker than ionic (or covalent) bonds. Average dipole - dipole interaction energies are approximately 4 kJ per mole of "bonds," whereas an average value for typical ionic and covalent bond energies is approximately 400 kJ per mol of bonds. The relatively high melting points and boiling points of many polar compounds can be attributed to dipole-dipole interactions.



Dipole - dipole interactions (---) in bromine fluoride and HCl molecules

QUESTIONS

1. Explain the formation of ionic, covalent and coordinate bonds with one example of each. Give briefly their important properties. P. U. 1983.
2. What is meant by hybridization of orbitals? Give the nature of hybridization in BF_3 , BeCl_2 and CCl_4 . P. U. 1983.
3. Discuss the principles involved in the valence shell electron pair repulsion concept. What are the applications of this theory in determining the shape of the molecules? P. U. 1987.
4. Discuss the principles involved in Valence Bond Theory. How is this theory applied to explain the formation of chemical bond?
5. Compare and contrast the Valence Bond and Molecular Orbital theories. P.U. 1990,2001
6. What do you understand by the term chemical bond? Discuss the formation of ionic and covalent bonds. P. U. 1989.
7. Distinguish between ionic and covalent bonds. P. U. 1990.
8. (a) Discuss "Molecular Orbital Theory" for the formation of the H_2 molecule.
(b) Draw molecular orbital diagram of O_2 molecule and discuss its paramagnetic nature. Non-existence of helium molecule on the basis of Molecular orbital theory. P. U. 1998,2000,2002
9. Describe the general theories of Metallic Bonding. P. U. 1984.
10. Predict the shapes of the following molecules on the basis of valence shell electron pair repulsion theory, (i) CH_4 (ii) NH_3 (iii) SnCl_2 (iv) PCl_5 (v) SF_6 (vi) BeCl_2 . P.U.1992, 1994.

11. Draw and explain the molecular structures of the following molecules.
(a) H_2 (b) O_2 (c) HF P.U. 1993
12. Name various theories which were put forward to explain the nature of metallic bonds and explain band theory. P. U. 1993, 1996
13. Discuss the effect of the following on the shapes of molecules with the help of suitable examples. (a) Lone pair of electrons (b) Multiple bonds P. U. 1993
14. Write notes on: (i) Metallic bond (ii) Valence bond theory. P. U. 1995
15. What is radius ratio? How are the value of limiting radius ratio able to predict the type of structures of ionic compounds. P. U. 1979
16. Explain fully with examples, what is meant by (i) σ bond (ii) π bond (iii) delocalization and (iv) hybridization. P. U. 1977
17. Predict the shape of the following molecules on the basis of valence shell electron pair repulsion theory. NH_4^+ , PCl_3 , $SnCl_2$, H_2O , IF_5 , H_2O . P.U. 1996.
- 18.(a) Discuss the molecular orbital theory. How is it applied to explain the formation of heteronuclear diatomic molecules?
(b) Explain the paramagnetic character of O_2 molecule based upon molecular orbital theory. P.U. 1999

SHORT QUESTIONS

Q.1. Why O_2 molecule is paramagnetic?

Ans. Molecular orbital configuration of O_2 is $KK(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p_x})^2(\pi_{2p_y})^2(\pi_{2p_z})^2(\pi_{2p_y}^*)^1(\pi_{2p_z}^*)^1$. As there two unpaired electrons the O_2 molecule is paramagnetic.

Q.2. Why N_2 molecule is diamagnetic?

Ans. Molecular orbital configuration of N_2 is $KK(\sigma_{2s})^2(\sigma_{2s}^*)^2(\sigma_{2p_x})^2(\pi_{2p_y})^2(\pi_{2p_z})^2$. As there is no unpaired electron. The molecule is diamagnetic.

Q.3. ClO_3^- and ClO_4^- ions have same number of electron pairs around central chlorine but their geometry is different. Why?

Ans. sp^3 hybridization takes place in the formation of both molecules. In ClO_3^- one tetrahedral position is occupied by a lone pair, therefore, ClO_3^- gets pyramidal shape. As excepted from sp^3 hybridization, ClO_4^- achieves tetrahedral shape.

Q.4. Indicate the hybrid orbital set used by the central atom in each of the following molecules and ions: (a) SiH_4 (b) CH_3^+ (c) ICl_2 (d) $MgCl_2$ (e) PF_6^- .

Ans. (a) sp^3 (b) sp^2 (c) sp^3d (d) ionic, hybrid orbitals not appropriate (e) sp^3d^2

Q.5. What is meant by the molecular geometry of a molecule (or ion)?

Ans. The molecular geometry of a molecule (or ion) is the arrangement of the atoms in space.

Q.6. Why is $lp - lp$ repulsion $> lp - bp$ repulsion $> bp - bp$ repulsion?

Ans. A bond pair is under the influence of two nuclei of bonding atoms whereas the lone pair is under the influence of only one nucleus. Hence lone pair is more active and mobile in causing repulsion with other orbitals. As a result $lp-lp$ repulsion is greater than $lp-bp$ repulsion which in turn is greater than $bp-bp$ repulsion.

Q.7. Indicate the species with sp^3 hybridization of central atom.

(a) SO_4^{2-} (b) $AsCl_3$ (c) NH_4^+ (d) ClO_2^- (e) BF_4^-

Ans. All are sp^3 hybridized.

Q.8. Indicate the hybridization and bond angles associated with each of the following geometries: (a) linear (b) tetrahedral (c) trigonal planar (d) octahedral (e) trigonal bipyramidal.

Ans. (a) sp , 180° (b) sp^3 , 109° (c) sp^2 , 120° (d) sp^3d^2 , 90° and 180° (e) sp^3d , 90° , 120° and 180° .

Q.9. What is meant by the term orbital overlap?

Ans. Orbital overlap occurs when valence atomic orbitals on two adjacent atoms share the same region of space.

Q.10. What is the significance of overlapping orbitals in valence bond theory?

Ans. In valence bond theory, orbital overlap allows two bonding electrons to mutually occupy the space between the bonded nuclei.

Q.11. What two fundamental concepts are incorporated in valence-bond theory?

Ans. Valence-bond theory is a combination of the atomic orbital concept and the Lewis model of electron-pair bonding.

Q.12. What hybrid orbitals would be expected for the central atom in each of the following: (a) BrF_5 , (b) BrF_3 (c) $AsCl_5$ (d) ClF_4^+

Ans. (a) sp^3d^2 (b) sp^3d (c) sp^3d (d) sp^3d

Q.13. Solid PCl_5 exists as $[PCl_4^+][PCl_6^-]$: Describe the electron structure of the PCl_6^- ion in terms of valence bond theory.

Ans. The P atom in PCl_6^- has six single bonds around it and not lone pairs. This suggests sp^3d^2 hybridization. Each bond is a σ (sigma) bond formed by overlap of an sp^3d^2 hybrid orbital on P with a $3p$ orbital on Cl and has octahedral structure.

Q.14. According to band theory, how do insulators differ from conductors? How do semiconductors differ from conductors?

Ans. According to band theory, an insulator has energy bands that are either

completely filled or completely empty bands, with a large energy gap between the full and empty bands. A conductor has partially filled energy bands. A *semiconductor* has a filled or partially filled energy band separated by a small energy gap from an empty or nearly empty band.

Q.15. Silver has the highest electrical and thermal conductivities of any metal. How does the electron-sea model account for these conductivities?

Ans. In the electron-sea model, electrons move about in the metallic lattice, while silver (metal) atoms remain more or less fixed in position. Under the influence of an applied potential, the electrons are free to move throughout the structure, giving rise to electrical and thermal conductivities.

Q.16. What is the difference between a localized π bond, and a delocalized one?

Ans. In a localized π bond, the electron density is concentrated between the two atoms forming the bond. In a delocalized π bond, the electron density is spread over all the atoms that contribute p orbitals to the net work.

Q.17. According to molecular-orbital theory, would Be_2 or Be_2^+ be expected to exist.

Ans. Molecular-orbital configuration of Be_2 is $\text{KK} (\sigma_{2s})^2 (\sigma_{2s}^*)^2$. The bond order is $\frac{1}{2}(2-2)=0$. No bond is formed, so Be_2 is not expected to exist and is not energetically favoured over isolated Be atoms. Molecular orbital configuration of Be_2^+ is $\text{KK} (\sigma_{2s})^2 (\sigma_{2s}^*)^1$. It has a bond order of $\frac{1}{2}(2-1)=0.5$ and is slightly lower in energy than isolated Be atoms. It will probably exist under special conditions, but be unstable.

Q.18. (a) How many electrons are there in the H_2^+ ion? Can you write a Lewis structure for the ion? Explain. (b) Write the electronic configuration of the ion in terms of its MOs. (c) What is the bond order?

Ans. (a) There is only one electron in H_2^+ , there is no bond by the Lewis definition, so it is not possible to write a Lewis structure, (b) $(\sigma_{1s})^1$ (c) Bond order = $\frac{1}{2}$.

Q.19. Which of the following ion would you expect to be paramagnetic: O_2^+ , N_2^{2-} , Li_2^+ , O_2^{2-} ?

Ans. Paramagnetic; O_2^+ , N_2^{2-} , Li_2^+ .

Q.20. What are the general conditions for the most effective combination of atomic orbitals in heteronuclear diatomic molecules?

Ans. The atomic orbitals involved should (i) have similar energies (ii) overlap as much as possible and (iii) have the same symmetry with respect to the internuclear axis.

Q.21. Define metallic bond.

Ans. The attractive force that binds the metal ions to the mobile, delocalized electrons is called metallic bond.

Q.22. What are the essential conditions to form a bond between two atoms

according to valence bond theory?

- Ans. 1. An orbital on one atom comes to occupy a portion of the same region of space as an orbital on the other atom. The two orbitals are said to overlap.
2. The total number of electrons in both orbitals is no more than two.

Q.23. What is meant by hybrid orbitals?

- Ans. Hybrid orbitals are orbitals used to describe bonding that are obtained by taking combinations of atomic orbitals of the isolated atoms.

Q.24. What is meant by bond order?

- Ans. The term bond order refers to the number of bonds that exist between two atoms. In molecular orbital theory, the bond order of a diatomic molecule is defined as one half the difference between the number of electrons in bonding orbitals, n_b , and the number of electrons in anti-bonding orbitals, n_a :

$$\text{Bond order} = \frac{1}{2}(n_b - n_a).$$

Q.25. What is molecular orbital theory?

- Ans. Molecular orbital theory is a theory of electronic structures of molecules in terms of molecular orbitals, which may spread over several atoms or the entire molecule. Each molecular orbital has a definite energy.

Q.26. Predict the shape or geometry of the following molecules, using VSEPR model: (a) CCl_4 (b) H_2Se (c) AsF_3 (d) AlCl_3

- Ans. (a) Tetrahedral (b) bent (c) trigonal pyramidal (d) trigonal planar

Q.27. Predict the geometries of the following ions, using the VSEPR model:

(a) SnCl_5^- (b) PF_6 (c) ClF_2^- (d) IF_4^-

- Ans. (a) trigonal bipyramidal (b) octahedral (c) linear (d) square planar

Q.28. What hybrid orbitals would be expected for the central atom in each of the following? (a) BrF_5 (b) BrF_3 (c) AsCl_5 (d) ClF_4^+

- Ans. (a) sp^3d^2 (b) sp^2d (c) sp^3d (d) sp^3d

Q.29. What is the molecular orbital configuration of HeH^+ . Do you expect the ion to be stable?

- Ans. $(\sigma_{1s})^2$, stable.

Q.30. What geometry is expected for the following molecules, according to VSEPR model? (a) PF_5 (b) BrF_3 (c) BrF_5 (d) SCl_4 .

- Ans. (a) trigonal bipyramidal (b) T-shaped (c) square pyramidal (d) distorted tetrahedral

Q.31. Describe the electronic structure of each of the following, using molecular orbital theory. Calculate the bond order of each and decide whether it should be stable. For each, state whether the substance is diamagnetic or paramagnetic: (a) B_2 (b) B_2^+ (d) O_2^-

- Ans. (a) $\text{KK}(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^2$, bond order = 1, stable, paramagnetic
(b) $\text{KK}(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^1$, bond order = 1/2, stable, paramagnetic
(c) $\text{KK}(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2(\pi_{2p}^*)^3$ bond order = 3/2, stable, paramagnetic.